

An exact gamma constant for an Apéry-like sequence

A279619: three independent proofs, all-orders asymptotics,
and a recessive companion

Research report prepared for Vladimir Reshetnikov

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Merged from two independent manuscripts of batch 73O2 of ProveIt's incoming-reports intake (manuscripts 58 and 56), with a third derivation from manuscript 32 of batch 77P5 (Sections 15 and 16)

$$a_n \sim \frac{\sqrt{3\pi}}{\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)} \frac{27^n}{n^{3/2}}.$$

The leading constant is evaluated exactly, the displayed OEIS asymptotic is proved, and the expansion is extended to arbitrary fixed order with a rigorous remainder. A rationally normalized companion exposes an exponentially smaller alternating scale.

Abstract. We study the level-seven Apéry-like sequence A279619, defined by a second-order polynomial recurrence. Its OEIS entry displays a conjectural asymptotic with a numerical constant. We prove that asymptotic and identify the constant as a reciprocal product of three gamma values. Three independent proofs of the constant are printed. The first: a rational parameter removes the square roots in the known hypergeometric generating function, making its singular connection coefficient explicit, and classical elliptic-period and Ramanujan identities reduce this coefficient to its gamma-product form. The second: the boundary value of the generating function is a Dedekind eta product at the Fricke fixed point $i/\sqrt{7}$, evaluated by the Chowla–Selberg formula at discriminant -7 , and the singular coefficient is then read off from Hirschhorn's constant for the square sequence A183204. The third, from a manuscript added later, differentiates the Fricke involution of the weight-one theta series at its fixed point to obtain the singular coefficient directly, and evaluates the same eta value from an Epstein zeta function with an explicit conductor-two factor. A functional equation gives a second exact sum, at $z = 1/125$. Frobenius expansion and singularity analysis give all algebraic correction terms, with rational coefficients and fixed-order error bounds, together with ratio, root and critical-tail expansions. We also construct a companion continued fraction, prove exact rational bracketing, and determine the amplitude and every correction term of its recessive solution. Applications include convolution powers, A183204, eventual log-convexity, and Lambert- W inversion with a separate treatment of integer rounding. Exact symbolic certificates and rational enclosures accompany numerical experiments. Classical inputs, previously calculated terms, sequence-specific deductions, and unsettled questions are explicitly distinguished.

Shipped with this report: verification programs of manuscripts 58 and 56 and the replay program of manuscript 32, their recorded outputs, rational certificates, numerical tables, and manuscript 58's source audit (listed in the report's README.md).

These are conventional mathematical proofs with computer-algebra checks, not Lean-certified theorems. Global priority is not claimed.

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1 The target and the scope of the contribution

The sequence begins

$$1, 2, 22, 336, 6006, 117348, 2428272, 52303680, 1160427510, \dots$$

and satisfies

$$(n+1)^2 a_{n+1} = (26n^2 + 13n + 2)a_n + 3(3n-1)(3n-2)a_{n-1}, \quad n \geq 1, \quad (1)$$

with $a_0 = 1, a_1 = 2$. (Manuscript 56 writes the last coefficient as $27n^2 - 27n + 6$.) Throughout, set

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad \rho = 1/27, \quad \mathcal{G}_7 = \Gamma(1/7)\Gamma(2/7)\Gamma(4/7). \quad (2)$$

This is A279619; $A(z)^2$ is the generating function of A183204 [1, 2]. Its connections with modular forms, hypergeometric functions, and another Apéry-like sequence make it a structured target rather than an isolated numerical curiosity.

1.1 What is already known

The OEIS entry consulted on October 1, 2026 labels its leading asymptotic as conjectural and supplies a numerical constant. O’Brien’s thesis [3, §8.4, pp. 63–67] obtained the recurrence and the level-seven modular parametrization, estimates that constant and formally calculates the first three correction terms. Those three terms are *not new here*. The hypergeometric representation is also recorded in OEIS in code attributed to Václav Kotešovec in 2018 [1]. We prove and rationalize that representation before using it analytically. The leading asymptotic of the square sequence A183204, with a complete expansion, is Hirschhorn’s theorem [4]; it is used, and its leading constant re-derived, in the second proof (Section 6).

The contribution is not a new general method of asymptotic analysis. It is an exact connection-constant evaluation, a rigorous all-orders fixed-order expansion, and an explicit recessive companion with a proved error amplitude. Standard machinery is used to close the displayed asymptotic question.

Item	Status in this report
Recurrence and hypergeometric function	Existing input; the latter receives a rational differential-equation certificate.
Leading scale and first three corrections	Previously conjectured or formally calculated; now given a remainder proof.
Gamma-product constant	Exact sequence-specific evaluation deduced from classical special values, by two independent routes.
Sum at $z = 1/125$	Deduced, in manuscript 56, from the constant and a functional equation.
Companion continued fraction	Constructed with exact bracketing and all-orders recessive asymptotics; global priority not asserted.
Convolution and inverse expansions	Deductions with explicit formulas and integer-rounding bounds; the inverse expansion is an instance of the repository’s inversion apparatus (Section 11).
Lucas congruences	Not resolved or used here.

A targeted search did not locate an earlier proof of this particular constant formula. This is not an exhaustive priority search. In particular, recent work of Cohen and Zudilin on continued fractions for Chowla–Selberg quotients [14] shows that the broader area is active. We do not claim a first continued fraction for a gamma quotient, an identification of the companion limit with such a quotient, or an irrationality measure. Manuscript 56’s own, independent audit is printed in Section 14.1.

1.2 Relation to ProveIt

The repository’s Analysis/Transseries project separates asymptotic scales, flatness, power–logarithmic expressions, and inversion from the Fabius-function development [15]. This report follows that separation: an analytic existence proof precedes exact coefficient recursions and inverse expansions. No unverified repository theorem is needed. The inspected snapshot was a5ad4ac2e2d4e642a6c015836a70492d40fcbc71; inspection was not an exhaustive review of archived reports.

[Merge note, batch 7302: The Lambert- W inverse of Section 11 is the instance $a = \log 27$, $\beta = -3/2$ of Theorem t2:thm:balanced-inverse of the repository’s volume *Combinatorial Transseries and Their Inverses* [16], which neither manuscript cites; it is printed as an application, with no novelty claimed for it. Manuscript 58’s rounding proposition is the sequence-specific part.]

1.3 Two sources, one report

This report merges two manuscripts that reached ProveIt on the same day, October 1, 2026, in batch 7302 of its incoming-reports intake, and that were written independently: manuscript 58 (*An exact gamma constant for an Apéry-like sequence*) and manuscript 56 (*Exact asymptotics of OEIS A279619*). Manuscript 58 inspected the repository at a commit made before manuscript 56 arrived, and manuscript 56 cites neither manuscript 58 nor any repository revision. Both prove the same leading constant, the same boundary value $A(\rho)$, and the same rational corrections (they agree as rationals through order 10, the last order manuscript 56 prints; manuscript 58 records them to order 12). They do so by *different* proofs. Two independent derivations of the same constant arriving on the same day is mild corroboration, not a substitute for review.

Manuscript 58 is the base text: Sections 1–5 and 7–12 are its text, with its labels prefixed, its hard-coded section numbers replaced by references, and manuscript 56’s additions inserted as marked passages. Manuscript 56 contributes the second proof of the constant (Section 6), the sum at $1/125$ and the critical-tail, ratio and root corollaries (Section 2), the integer form of the Frobenius recurrence and the coefficients $c_6, \dots, c_{10}$ (Section 7), its inverse remark (Section 11), its numerical tables (Section 12), ten research problems (Section 13.13), its novelty audit (Section 14.1), two derivations (Appendix B) and its OEIS proposal (Appendix C). Results proved in both manuscripts are printed once and marked “proved in both sources” (for Theorem 2.1, now “proved in all three sources”). Section 15 records the provenance.

[Write note, 2026-10-02, batch 77P5: A third manuscript, manuscript 32 of batch 77P5 (*The exact asymptotic constant for A279619 and its fixed weight family*, dated October 2, 2026), reached ProveIt a day later. It names no repository revision and cites neither this report nor manuscripts 56 and 58, and it proves Theorem 2.1 a third time, by a third route: the singular coefficient directly at the Fricke fixed point, and the eta value by an explicit Epstein-zeta calculation. It contains no theorem that is not already in this report. Its route is printed in Section 16, placed after the provenance section so that no existing section, theorem or equation number changes, with its own notation table (Table 7); its explicit convolution generator and its all-weight rounding brackets are Remarks 10.2 and 11.3.]

1.4 Notation of the two manuscripts

The two manuscripts use several letters for different objects. This report uses manuscript 58's notation throughout; manuscript 56's material has been rewritten in it. The dictionary, with the tempting false readings, is Table 1.

Table 1: Notation of manuscripts 58 and 56.

Object	This report (= 58)	Manuscript 56	Caution
the sequence A279619	a_n	c_n	here c_j are <i>corrections</i>
its generating function	$A(z)$	$F(x)$	here F is ${}_2F_1(\frac{1}{12}, \frac{5}{12}; 1; \cdot)$
gamma product	$\mathcal{G}_7$	$\mathcal{G}_7$	same
leading constant	C	C_c	—
boundary value $A(\rho)$	$A_\star$	$F(1/27), A(0)$	56's $A(s)$ is not the GF
corrections	c_j	α_j	equal as rationals
analytic part at ρ	$U(s)$	$A(s)$	—
singular coefficient	B	B_0	—
singular Frobenius series	$V(s) = \sum v_j s^j$	$B(s) = \sum b_m s^m$	here b_n is the <i>companion</i>
square sequence (A183204)	$t_n = a_n^{(2)}, A(z)^2$	$t_n, G(x) = F(x)^2$	G is not $\mathcal{G}_7$
its leading constant	$C_t = 2CA_\star$	C_t	—
$\log 27$	λ	L	here L is the companion limit
entropy function	$\text{Ent}(u)$	$H(u)$	here H is Clausen's ${}_3F_2$
Laplace variable, weight	$\xi = k/n, \kappa(\xi)$	$a, g(a)$	a is not a term
scaled term	$R_n = a_n n^{3/2}/27^n$	Q_n	here $Q_n, Q(m)$ differ
critical partial sum, shift	$\Sigma_N, \nu = N + 1$	S_N, q	here S_N sums $F(q_0)$
continuous inverse	$x_0(Y)$	$n_0(y)$	—
formal series at infinity	$S_{27}(t)$	$\mathcal{A}(t)$	—
Weber invariant power	$u = f_2(\tau_0)^{24}$	u	here u_n is a solution
level-7 Hauptmodul	$w_7(\tau)$	$w_7(\tau)$	here w is the rational parameter
modular j -invariant	$j(\tau)$	$j(\tau)$	here $j(m)$ is a pullback

No normalization changes: $C = C_c, B = B_0$ (both with the singular Frobenius series normalized to 1 at $s = 0$) and $\lambda = L_{56}$. Manuscript 56's index range “ $M \geq 1$, error $O(n^{-M})$, sum to $M - 1$ ” is manuscript 58's “ $M \geq 0$, error $O(n^{-M-1})$, sum to M ” shifted by one.

2 Principal results

Theorem 2.1 (Dominant asymptotics and the exact constant; proved in all three sources). *Put*

$$C = \frac{\sqrt{3\pi}}{\mathcal{G}_7}, \quad A_\star = \frac{3\mathcal{G}_7}{8\pi^2}. \quad (3)$$

Then $A(\rho) = A_\star$. For every fixed $M \geq 0$,

$$a_n = C \frac{27^n}{n^{3/2}} \left(\sum_{j=0}^M \frac{c_j}{n^j} + O_M(n^{-M-1}) \right), \quad (4)$$

where the c_j are uniquely determined rational numbers with $c_0 = 1$. In particular,

$$\begin{aligned} c_1 &= -\frac{215}{1008}, & c_2 &= -\frac{1265}{290304}, & c_3 &= \frac{4683055}{877879296}, \\ c_4 &= \frac{924908071}{3539609321472}, & c_5 &= -\frac{6088275239005}{3567926196043776}. \end{aligned} \quad (5)$$

An exact triangular algorithm computes every coefficient.

The theorem also evaluates a convergent boundary sum:

$$\sum_{n=0}^{\infty} \frac{a_n}{27^n} = \frac{3\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)}{8\pi^2}, \quad CA_{\star} = \frac{3\sqrt{3}}{8\pi^{3/2}}. \quad (6)$$

The gamma product in the boundary value cancels against its reciprocal in the coefficient amplitude. Numerically $C = 0.0955223052681267146513079107870296256727946666510071798669\dots$ and $A_{\star} = 1.221131097467566669284224585685993462375827056379\dots$; the two manuscripts' decimal values agree in every printed digit.

[Merge note, batch 7302: Manuscript 58 proves Theorem 2.1 in Sections 3–7; manuscript 56 proves it in Section 6 and in its part of Section 7. Manuscript 56's statement is the same theorem with $\alpha_j = c_j$ and the shifted index range of Section 1.4; it lists the same five coefficients.] [Write note, 2026-10-02, batch 77P5: Manuscript 32 proves the theorem a third time (Section 16), with $p_m = c_m$ in its notation; it prints $c_1, \dots, c_6$, equal as rationals to (5) and (76).]

Theorem 2.2 (Exact critical and subcritical sums; manuscript 56). *The generating function has the exact values*

$$A(1/27) = \frac{3\mathcal{G}_7}{8\pi^2}, \quad A(1/125) = \sum_{n=0}^{\infty} \frac{a_n}{125^n} = \frac{5\mathcal{G}_7}{16\pi^2}. \quad (7)$$

Numerically, $A(1/125) = 1.017609247889638891070187154738327885313189213649\dots$

The first value is (6). The second is proved in Section 6.6 from the first and a functional equation; it appears only in manuscript 56.

What is resolved (manuscript 56). The decimal constant in the OEIS asymptotic conjecture is exactly the gamma quotient in (3). More is true: the complete expansion exists to arbitrary algebraic order, all its normalized coefficients are rational, the singular connection coefficient is explicit, and the critical value of the generating function is a closed gamma product.

2.1 Immediate corollaries

The all-orders form gives several useful consequences without further modular input. The three corollaries below are manuscript 56's.

Corollary 2.3 (Ratio asymptotics and eventual log-convexity). *As $n \rightarrow \infty$,*

$$\frac{a_{n+1}}{a_n} = 27 \left(1 - \frac{3}{2n} + \frac{2105}{1008n^2} - \frac{18815}{7056n^3} + \frac{3425255}{1053696n^4} + O(n^{-5}) \right). \quad (8)$$

Consequently $a_{n+1}/a_n \rightarrow 27$, and (a_n) is strictly log-convex for all sufficiently large n .

Proof. Insert (4) into the quotient. The difference of successive ratios is

$$\frac{a_{n+2}}{a_{n+1}} - \frac{a_{n+1}}{a_n} = \frac{81}{2n^2} + O(n^{-3}) > 0$$

for all sufficiently large n . Increasing ratios are equivalent to strict log-convexity. $\square$

Manuscript 58 proves the first three terms of (8) and eventual log-convexity through second differences of $\log a_n$ (Section 7.4); the two increments are consistent, $81/(2n^2) \div 27 = 3/(2n^2)$.

Corollary 2.4 (Root and logarithmic scales). *With $\lambda = \log 27$,*

$$\log a_n = n\lambda - \frac{3}{2} \log n + \log C - \frac{215}{1008n} + O(n^{-2}), \quad (9)$$

$$a_n^{1/n} = 27 \exp\left(\frac{\log C - \frac{3}{2} \log n}{n} + O(n^{-2})\right). \quad (10)$$

The first line is the case $M = 1$ of manuscript 58's logarithmic expansion (77).

Corollary 2.5 (Critical-tail asymptotics). *Let $\Sigma_N = \sum_{n=0}^N a_n/27^n$ and $\nu = N + 1$. Then*

$$A(1/27) - \Sigma_N = C \left(2\nu^{-1/2} + \frac{541}{1512} \nu^{-3/2} + \frac{2411}{145152} \nu^{-5/2} + O(\nu^{-7/2}) \right). \quad (11)$$

In particular, convergence at the critical point is only of order $N^{-1/2}$.

Proof. Sum (4) termwise and use the standard Hurwitz-zeta expansion for $\zeta(s, N + 1)$. $\square$

[Merge note, batch 7302: The intake re-derived 541/1512 and 2411/145152 by the Euler–Maclaurin formula, and checked that the partial sum through $N = 5000$ plus the three-term tail matches $3\mathcal{G}_7/(8\pi^2)$ to 4.9×10^{-16} , consistent with the stated $O(\nu^{-7/2})$.]

Theorem 2.6 (An exact recessive companion; manuscript 58). *Let $b_0 = 0$, $b_1 = 1$, and impose (1) on b_n for $n \geq 1$. Then the limit*

$$L = \lim_{n \rightarrow \infty} \frac{b_n}{a_n} = 0.46721176888437312394211005141309126172410854995971 \dots \quad (12)$$

exists and satisfies

$$\frac{b_{2m}}{a_{2m}} < L < \frac{b_{2m+1}}{a_{2m+1}}, \quad m \geq 0. \quad (13)$$

For $e_n = b_n - La_n$ and every fixed $M \geq 0$,

$$e_n = (-1)^{n+1} \frac{\sqrt{3}}{56\pi C} n^{-3/2} \left(\sum_{j=0}^M \frac{d_j}{n^j} + O_M(n^{-M-1}) \right), \quad (14)$$

where $d_0 = 1$ and

$$d_1 = -\frac{57}{112}, \quad d_2 = \frac{655}{3584}, \quad d_3 = -\frac{13365}{401408}, \quad d_4 = -\frac{3627123}{179830784}. \quad (15)$$

The d_j are rational and computable to arbitrary order. Consequently,

$$L - \frac{b_n}{a_n} \sim (-1)^n \frac{\sqrt{3}}{56\pi C^2} 27^{-n}. \quad (16)$$

The two scales are $27^n n^{-3/2}$ and $(-1)^n n^{-3/2}$. The second is exponentially small *relative* to the first, not in absolute magnitude. We do not assign an unproved canonical subdominant coefficient to the asymptotic expansion of a_n itself.

3 From the recurrence to a controlled singularity

3.1 Positivity, growth, and the differential equation

The recurrence immediately gives $a_n > 0$. Induction also gives $a_n \leq 27^n$, since

$$26n^2 + 13n + 2 + \frac{3(3n-1)(3n-2)}{27} = 27n^2 + 12n + \frac{20}{9} < 27(n+1)^2.$$

Thus A is analytic in $|z| < \rho$. Multiplying the recurrence by z^n and summing, including $a_1 = 2a_0$, gives

$$zD(z)A'' + (1 - 39z - 54z^2)A' - (2 + 6z)A = 0, \quad D(z) = (1 - 27z)(1 + z). \quad (17)$$

The coefficient of z^n on the left is the recurrence residual, so the analytic solution normalized by $A(0) = 1$ is unique. Outside $0, \rho, -1$, the equation is ordinary; analytic continuation along paths avoiding these points follows from linear differential-equation theory. (Manuscript 56 derives the same equation, written with $zD(z) = z(1 - 26z - 27z^2)$, in Appendix B. Its positive singularity ρ is separated from the other nonzero finite singularity -1 by a factor of 27 in modulus.)

3.2 The local exponents

In $s = 1 - 27z$, the equation becomes

$$s(1-s)(28-s)y'' + (14 - 43s + 2s^2)y' - \frac{20-2s}{9}y = 0. \quad (18)$$

The indicial equation is $14\alpha(2\alpha - 1) = 0$. Its exponents $0, 1/2$ have nonintegral difference, hence Frobenius theory gives a convergent local expansion

$$A(z) = U(s) + B\sqrt{s}V(s), \quad V(0) = 1, \quad (19)$$

where U, V are analytic at 0. Branches are positive for $0 < z < \rho$. Below we show $B < 0$, so the singularity is not removable.

The self-adjoint form provides a useful sign check:

$$(z\sqrt{D(z)}A'(z))' = \frac{2+6z}{\sqrt{D(z)}}A(z), \quad 0 < z < \rho. \quad (20)$$

Frobenius theory makes $A(\rho)$ finite. Positivity of the right side implies that the limiting flux is positive, whereas (19) identifies it as $-\frac{1}{2}\sqrt{28/27}B$.

There is no other singularity on $|z| = \rho$. The chosen solution extends to a dented disk of radius greater than ρ , cut to the right from ρ . This supplies the global analytic condition for singularity analysis, not just a formal local expansion [12, Chapter VI].

4 A rational certificate for the generating function

Set

$$w = \sqrt{\frac{1-27z}{1+z}}, \quad z = \frac{1-w^2}{27+w^2}. \quad (21)$$

The points $0, \rho$ correspond to $w = 1, 0$. Define

$$\begin{aligned} T(w) &= 15w^2 + 96w + 85, \\ P(w) &= \left(\frac{7(w^2 + 27)}{T(w)} \right)^{1/4}, \\ J(w) &= \frac{1728(1-w)^7(1+w)}{(w^2 + 27)T(w)^3}, \quad F(x) = {}_2F_1 \left(\begin{matrix} 1/12, 5/12 \\ 1 \end{matrix}; x \right). \end{aligned} \quad (22)$$

Fourth roots are positive on $[0, 1]$.

Proposition 4.1. *As analytic germs near $w = 1$, and throughout $0 \leq w \leq 1$,*

$$A \left(\frac{1-w^2}{27+w^2} \right) = P(w)F(J(w)). \quad (23)$$

Proof. The equation for $y(w) = A(z(w))$ is $y'' + py' + qy = 0$, where

$$p = \frac{2w}{(w-1)(w+1)}, \quad q = -\frac{16(w^2-15)}{(w-1)(w+1)(w^2+27)^2}. \quad (24)$$

Let $h = P'/P = \frac{1}{4}(2w/(w^2+27) - T'/T)$. The hypergeometric equation [13, §15.10] is

$$x(1-x)F'' + (1-\frac{3}{2}x)F' - \frac{5}{144}F = 0. \quad (25)$$

Substitution of $y = PF(J)$, followed by eliminating F'' , reduces the claim to two rational identities:

$$\begin{aligned} J'' + (2h+p)J' + (J')^2 \frac{3J/2-1}{J(1-J)} &= 0, \\ h' + h^2 + ph + q + \frac{5}{144} \frac{(J')^2}{J(1-J)} &= 0. \end{aligned} \quad (26)$$

Clearing denominators and expanding proves both; the attached code checks them identically in $\mathbb{Q}(w)$. This is an exact certificate rather than a fit to finitely many coefficients. At $w = 1$, $P = 1$, $J = 0$, and $z'(1) \neq 0$. The right side is analytic there and normalized to 1. Uniqueness at $z = 0$ proves the germ identity. For $0 \leq w < 1$,

$$\frac{J'}{J} = -\frac{7}{1-w} + \frac{1}{1+w} - \frac{2w}{w^2+27} - 3\frac{30w+96}{T(w)} < 0.$$

Thus $0 \leq J(w) \leq J(0) = 64/85^3 < 1$, and every expression continues regularly along the interval, proving the full claim. $\square$

Differentiation at a singular z -value has now become ordinary differentiation at the regular parameter value $w = 0$.

5 Evaluating the connection constant (first proof)

5.1 An explicit hypergeometric expression

Put

$$q_0 = \frac{64}{85^3}, \quad P_0 = \left(\frac{189}{85} \right)^{1/4}. \quad (27)$$

As $s \rightarrow 0$,

$$w = \sqrt{\frac{27s}{28-s}} = \sqrt{\frac{27}{28}}\sqrt{s}(1 + O(s)).$$

At $w = 0$ we have $P'/P_0 = -24/85$ and $J'/q_0 = -798/85$. Hence

$$B = -\sqrt{\frac{27}{28}} \frac{P_0}{85} (24F(q_0) + 798q_0F'(q_0)) < 0. \quad (28)$$

Positivity follows from the positive hypergeometric coefficients. Coefficient transfer for $\sqrt{1-27z}$ and $\Gamma(-1/2) = -2\sqrt{\pi}$ gives

$$C = -\frac{B}{2\sqrt{\pi}} = \sqrt{\frac{27}{28}} \frac{P_0}{170\sqrt{\pi}} (24F(q_0) + 798q_0F'(q_0)). \quad (29)$$

This already gives an explicit positive constant. The following special-value reductions explain its unexpectedly simple form.

5.2 Ramanujan's identity removes the derivative

The relevant Clausen identity is

$$F(q)^2 = H(q) := \sum_{n \geq 0} \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} q^n. \quad (30)$$

It follows directly from differential equations: the symmetric square of (25) is

$$[\vartheta^3 - q(\vartheta + 1/6)(\vartheta + 1/2)(\vartheta + 5/6)]H = 0, \quad \vartheta = q \frac{d}{dq},$$

whose analytic solution with $H(0) = 1$ is unique. The rational coefficients of this symmetric-square identity are checked in the code.

The classical level-seven Ramanujan identity, reproduced and obtained by the construction in [6, Eq. (98)], is

$$\sum_{n \geq 0} (133n + 8) \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} q_0^n = \frac{85\sqrt{255}}{54\pi}. \quad (31)$$

The source uses the coefficient $(6n)!/((3n)!(n!)^3)$, which is 1728^n times the coefficient here, and $1728/255^3 = 64/85^3$. This normalization check is essential. Equation (31) is a cited established identity, not an equality inferred from numerical recognition. Using (30), it becomes

$$8F(q_0)^2 + 266q_0F(q_0)F'(q_0) = \frac{85\sqrt{255}}{54\pi}.$$

Multiply by $3/F(q_0)$ and substitute into (29). Since $A_\star = P_0F(q_0)$ by Proposition 4.1, this yields

$$C = \frac{3\sqrt{3}}{8\pi^{3/2}P_0F(q_0)} = \frac{3\sqrt{3}}{8\pi^{3/2}A_\star}. \quad (32)$$

5.3 An elliptic period supplies the gamma product

Let

$$m_7 = \frac{8 - 3\sqrt{7}}{16}, \quad k_7 = \sqrt{m_7}, \quad K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}. \quad (33)$$

Here K takes the *modulus* k , not the parameter $m = k^2$. The classical singular-period evaluation is

$$K(k_7) = \frac{\mathcal{G}_7}{4\pi 7^{1/4}}. \quad (34)$$

This is an established complex-multiplication special value [7, 8], not a new identity claimed here. Appendix A explicitly reduces the beta-function form in Zucker’s table to this normalization.

The required hypergeometric transformation is

$$F\left(\frac{27m^2(1-m)^2}{4(1-m+m^2)^3}\right) = (1-m+m^2)^{1/4} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; m\right). \quad (35)$$

For a direct verification put $Q(m) = 1 - m + m^2$ and $j(m) = 27m^2(1-m)^2/(4Q(m)^3)$. Then $Q^{-1/4}F(j)$ satisfies

$$m(1-m)y'' + (1-2m)y' - \frac{1}{4}y = 0.$$

Substituting and clearing denominators proves two rational identities, exactly as in (26). Both solutions are analytic at 0 and equal to 1, proving the transformation near 0. Continuation on $[0, m_7]$ proves the value used here. The program verifies both identities.

At $m = m_7$,

$$m_7(1-m_7) = 1/256, \quad Q(m_7) = 255/256, \quad j(m_7) = q_0.$$

Euler’s integral gives ${}_2F_1(1/2, 1/2; 1; m_7) = 2K(k_7)/\pi$. Therefore

$$\begin{aligned} A_\star &= P_0 F(q_0) = \left(\frac{189}{85} \frac{255}{256}\right)^{1/4} \frac{2K(k_7)}{\pi} \\ &= \frac{3 \cdot 7^{1/4}}{4} \frac{2}{\pi} \frac{\mathcal{G}_7}{4\pi 7^{1/4}} = \frac{3\mathcal{G}_7}{8\pi^2}. \end{aligned}$$

Combining this with (32) proves $C = \sqrt{3\pi}/\mathcal{G}_7$. Since the coefficients are positive, the finite limit at ρ also gives the boundary sum (6), by monotone convergence.

Remark 5.1 (Dependencies of the exact evaluation). The two non-elementary special values are (31) and (34); both are classical inputs. The recurrence manipulations, rational pullbacks, connection-coefficient extraction, and asymptotic deductions are proved here. We do not purport to give a new derivation of Chowla–Selberg theory or of Ramanujan’s general $1/\pi$ formulas.

6 Second proof: the square sequence and a CM evaluation (manuscript 56)

This section is manuscript 56’s proof of $A_\star = 3\mathcal{G}_7/(8\pi^2)$ and $C = \sqrt{3\pi}/\mathcal{G}_7$, rewritten in this report’s notation (Section 1.4). It is independent of Sections 4 and 5: it uses neither the ${}_2F_1$ pullback, nor Ramanujan’s series, nor the elliptic period. Its classical inputs are O’Brien’s modular parametrization, the Chowla–Selberg formula at discriminant -7 , and Hirschhorn’s asymptotic for A183204, whose leading constant it re-derives.

The proof combines four structures already present around this level-seven sequence: the square identity $A(z)^2 = \sum t_n z^n$ with the Apéry-like sequence A183204, a one-dimensional Laplace analysis of the binomial sum for t_n , O'Brien's modular parametrization, and a Chowla–Selberg evaluation of a Dedekind eta value of discriminant -7 . They come from two apparently different descriptions of the same object: a square root of an Apéry-like generating function and a modular parametrization on $X_0(7)$.

6.1 The square identity

Let

$$A(z)^2 = \sum_{n \geq 0} t_n z^n, \quad t_n = a_n^{(2)}. \quad (36)$$

O'Brien's level-seven construction, equivalently the relation recorded in OEIS, identifies (t_n) with A183204, one of the sporadic Apéry-like sequences [9]. (Manuscript 58 checks the identification through the symmetric-square equation (93).) Its coefficients have the finite binomial sum

$$t_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{2k}{n} \binom{n+k}{k}, \quad (37)$$

where terms with $2k < n$ vanish. The initial values are

$$1, 4, 48, 760, 13840, 273504, \dots$$

[Merge note, batch 7302: Manuscript 58 checks the convolution against the different binomial form $\sum_k \binom{n}{k}^2 \binom{n+k}{n} \binom{2n-k}{n}$; the intake checked that both forms equal $[z^n]A(z)^2$ for $n \leq 40$.] Hirschhorn proved a complete asymptotic expansion for this sequence [4]. We rederive its leading constant in Section 6.3, because that constant is the bridge from the known square to the previously unknown square root.

6.2 The modular parametrization

Let η denote Dedekind's eta function [13, Ch. 23],

$$\eta(\tau) = e^{\pi i \tau / 12} \prod_{m \geq 1} (1 - e^{2\pi i m \tau}), \quad \Im \tau > 0. \quad (38)$$

O'Brien's parametrization may be written

$$w_7(\tau) = \left(\frac{\eta(7\tau)}{\eta(\tau)} \right)^4, \quad (39)$$

$$X_7(\tau) = \frac{w_7(\tau)}{1 + 13w_7(\tau) + 49w_7(\tau)^2}, \quad (40)$$

$$X_7(\tau) = \left(\frac{\eta(\tau)\eta(7\tau)}{z_7(\tau)} \right)^3, \quad A(X_7(\tau)) = z_7(\tau). \quad (41)$$

The last identity is initially a q -series identity near the cusp and then extends by analytic continuation. It is the crucial reason the critical value of A can be reduced to a CM eta product.

6.3 The square sequence by Laplace's method

We now obtain the leading asymptotic of t_n directly from (37). This gives a self-contained derivation of the one published constant from which the new constant will be extracted.

For $\xi \in (1/2, 1)$ and $k = \xi n$, Stirling's formula gives, uniformly on closed subintervals of $(1/2, 1)$,

$$\binom{n}{k}^2 \binom{2k}{n} \binom{n+k}{k} = n^{-2} \kappa(\xi) e^{n\Phi(\xi)} (1 + O(n^{-1})), \quad (42)$$

where, with $\text{Ent}(u) = -u \log u - (1-u) \log(1-u)$,

$$\Phi(\xi) = 2 \text{Ent}(\xi) + 2\xi \text{Ent}\left(\frac{1}{2\xi}\right) + (1+\xi) \text{Ent}\left(\frac{\xi}{1+\xi}\right), \quad (43)$$

$$\kappa(\xi) = \frac{1}{2\pi^2 \xi(1-\xi)} \sqrt{\frac{1+\xi}{2(2\xi-1)}}. \quad (44)$$

A convenient derivative is

$$\Phi'(\xi) = 2 \log(1-\xi) + \log\left(4 + \frac{4}{\xi}\right) - 2 \log(2\xi-1). \quad (45)$$

Furthermore,

$$\Phi''(\xi) = -\frac{2}{1-\xi} - \frac{1}{\xi(1+\xi)} - \frac{4}{2\xi-1} < 0,$$

so the critical point is the unique global maximum. Direct substitution gives

$$\xi_0 = \frac{4}{5}, \quad e^{\Phi(\xi_0)} = 27, \quad \Phi''(\xi_0) = -\frac{625}{36}, \quad \kappa(\xi_0) = \frac{25\sqrt{6}}{16\pi^2}. \quad (46)$$

The endpoint ranges are exponentially smaller than the saddle range. Replacing the sum by a Riemann sum and applying the Gaussian form of Laplace's method therefore yields

$$\begin{aligned} t_n &\sim 27^n n^{-3/2} \kappa(\xi_0) \sqrt{\frac{2\pi}{-\Phi''(\xi_0)}} \\ &= C_t 27^n n^{-3/2}, \quad C_t = \frac{3\sqrt{3}}{4\pi^{3/2}} = \frac{1}{4} \left(\frac{3}{\pi}\right)^{3/2}. \end{aligned} \quad (47)$$

This agrees with Hirschhorn's theorem, whose further terms begin

$$t_n \sim C_t 27^n n^{-3/2} \left(1 - \frac{65}{144n} + \frac{3865}{41472n^2} + \frac{111727}{17915904n^3} + \cdots\right). \quad (48)$$

The first correction $-65/144$ is also the case $m = 2$ of manuscript 58's convolution theorem, (92); the two manuscripts agree.

Remark 6.1 (Why the square alone does not determine the answer). From $A^2 = \sum t_n z^n$, the asymptotic constant of t_n is not simply the square of the constant of a_n . Both series converge at $z = \rho$, and their dominant singularity is additive rather than multiplicative: $A = A(\rho) + B\sqrt{1-z/\rho} + \cdots$. Thus the singular coefficient of A^2 is $2A(\rho)B$. The missing datum is the exact boundary value $A(\rho)$, which is evaluated next.

6.4 The Fricke fixed point

Take

$$\tau_* = \frac{i}{\sqrt{7}}. \quad (49)$$

Dedekind's inversion law [11, Ch. 3]

$$\eta(-1/\tau) = \sqrt{-i\tau} \eta(\tau) \quad (50)$$

shows that

$$\eta(i\sqrt{7}) = 7^{-1/4} \eta(i/\sqrt{7}). \quad (51)$$

It follows from (39) that $w_7(\tau_*) = 1/7$, hence

$$X_7(\tau_*) = \frac{1/7}{1 + 13/7 + 1} = \frac{1}{27} = \rho. \quad (52)$$

All relevant quantities are positive on the imaginary axis. Therefore (41) gives

$$A(\rho) = z_7(\tau_*) = 3\eta(i/\sqrt{7})\eta(i\sqrt{7}). \quad (53)$$

The remaining task is an exact evaluation of this product.

6.5 Chowla–Selberg at discriminant -7

Put

$$\tau_0 = \frac{1 + i\sqrt{7}}{2}. \quad (54)$$

The class number of discriminant -7 is one [10]. Specializing the Chowla–Selberg formula in the form proved by van der Poorten and Williams [5] gives

$$|\eta(\tau_0)| = (14\pi)^{-1/4} \left(\frac{\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)}{\Gamma(3/7)\Gamma(5/7)\Gamma(6/7)} \right)^{1/4}. \quad (55)$$

Reflection [13, Ch. 5] and the classical trigonometric product

$$\sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{8} \quad (56)$$

imply

$$\mathcal{G}_7 \Gamma(3/7)\Gamma(5/7)\Gamma(6/7) = \frac{8\pi^3}{\sqrt{7}}.$$

Consequently (55) simplifies to

$$|\eta(\tau_0)| = \frac{\sqrt{\mathcal{G}_7}}{2 \cdot 7^{1/8} \pi}. \quad (57)$$

We still need to move from τ_0 to $i\sqrt{7}$. Introduce the Weber function

$$\mathfrak{f}_2(\tau) = \sqrt{2} \frac{\eta(2\tau)}{\eta(\tau)}, \quad j(\tau) = \frac{(\mathfrak{f}_2(\tau)^{24} + 16)^3}{\mathfrak{f}_2(\tau)^{24}}. \quad (58)$$

The Hilbert class polynomial of discriminant -7 is $X + 3375$ [10], so $j(\tau_0) = -3375$. Setting $u = \mathfrak{f}_2(\tau_0)^{24}$ gives

$$(u + 16)^3 + 3375u = (u + 1)(u^2 + 47u + 4096) = 0. \quad (59)$$

Here the nome is $e^{2\pi i \tau_0} = -e^{-\pi\sqrt{7}}$, and the product definition of η shows that u is a negative real number. The two roots of the quadratic factor in (59) are nonreal. Hence $u = -1$ and $|\mathfrak{f}_2(\tau_0)| = 1$. Since $2\tau_0 = 1 + i\sqrt{7}$ and $|\eta(\tau + 1)| = |\eta(\tau)|$, we obtain

$$\eta(i\sqrt{7}) = \frac{|\eta(\tau_0)|}{\sqrt{2}} = \frac{\sqrt{\mathcal{G}_7}}{2\sqrt{2}7^{1/8}\pi}. \quad (60)$$

Combining (51) and (60),

$$\eta(i/\sqrt{7})\eta(i\sqrt{7}) = 7^{1/4}\eta(i\sqrt{7})^2 = \frac{\mathcal{G}_7}{8\pi^2}. \quad (61)$$

Finally, (53) becomes

$$A(1/27) = \frac{3\mathcal{G}_7}{8\pi^2}. \quad (62)$$

This is the second proof of $A(\rho) = A_\star$ in Theorem 2.1.

6.6 A second exact series from the functional equation

The square-root generating function satisfies

$$\frac{1}{1+4y}A\left(\frac{y}{(1+4y)^3}\right) = \frac{1}{1+2y}A\left(\frac{y^2}{(1+2y)^3}\right). \quad (63)$$

This follows either from O'Brien's modular equation or by taking the branch equal to 1 at the origin in the corresponding identity for A^2 . At $y = 1/8$, the two arguments are $1/27$ and $1/125$, and (63) reads

$$\frac{2}{3}A(1/27) = \frac{4}{5}A(1/125).$$

Together with (62), this proves the second identity of Theorem 2.2.

[Merge note, batch 7302: Provenance of (63). Manuscript 56 derives it from O'Brien's modular equation (or from the identity for A^2) and uses it only here. Manuscript 58 mentions, without using it, a functional equation added to the OEIS entry in 2026 (August 2026, in its source audit) as a possible input for the Lucas congruences (Section 13.6); the dossier of the intake identifies it with (63). The intake checked (63) as an identity of power series in y through y^{24} , and the value $A(1/125) = 5\mathcal{G}_7/(16\pi^2)$ by direct summation, to 3×10^{-61} . The second identity of Theorem 2.2 therefore rests on the functional equation, taken as an input from O'Brien's modular equation (the finite check is not a proof), as a power-series identity continued along $0 \leq y \leq 1/8$, where both arguments increase within $[0, \rho]$ and reach $1/27$ and $1/125$ at $y = 1/8$.]

6.7 The singular connection coefficient

Because the exponents at $s = 0$ are 0 and $1/2$, the local form of A is (19), $A(z) = U(s) + B\sqrt{s}V(s)$, with $U(0) = A(\rho)$ and $V(0) = 1$. The coefficient B is nonzero: otherwise A , and hence A^2 , would be analytic at $1/27$, contradicting (47).

Squaring (19),

$$A(z)^2 = A(\rho)^2 + 2A(\rho)Bs^{1/2} + O(s). \quad (64)$$

The transfer formula (73) at exponent $1/2$ [12, Ch. VI],

$$[z^n](1 - 27z)^{1/2} \sim -\frac{27^n}{2\sqrt{\pi}n^{3/2}}, \quad (65)$$

then shows that the constant in (47) satisfies

$$C_t = -\frac{2A(\rho)B}{2\sqrt{\pi}} = -\frac{A(\rho)B}{\sqrt{\pi}}. \quad (66)$$

(The transfer from the local expansion of A^2 to t_n uses the same dented-disk continuation as in Section 3; see the merge note in Section 7.2.) Using (47) and (62),

$$B = -\frac{2\pi\sqrt{3}}{\mathcal{G}_7}. \quad (67)$$

Applying (65) directly to A gives

$$C = -\frac{B}{2\sqrt{\pi}} = \frac{\sqrt{3\pi}}{\mathcal{G}_7},$$

which is the second proof of the leading assertion of Theorem 2.1. In manuscript 58's terms, (66) is $C_t = 2CA_\star$, the case $m = 2$ of (90); manuscript 56 obtains the product $CA_\star$ from the known square, manuscript 58 from Ramanujan's series (32).

Exact local connection formula (manuscript 56). At the dominant singularity,

$$A(z) = \frac{3\mathcal{G}_7}{8\pi^2} - \frac{2\pi\sqrt{3}}{\mathcal{G}_7}\sqrt{1-27z} + O(1-27z). \quad (68)$$

The first constant is modular/CM; the second is obtained by matching the known square A^2 to the first. Formula (3) is precisely the transfer of the second constant.

Manuscript 58 obtains the same singular coefficient in hypergeometric form, (28), and the same value through $B = -2\sqrt{\pi}C$.

[Write note, 2026-10-02, batch 77P5: Manuscript 32 obtains B a third way, without the square sequence and without a hypergeometric identity: by differentiating the Fricke involution of z_7 at τ_* (Proposition 16.2, $BA_\star = -3\sqrt{3}/(4\pi)$). It also re-derives (60) from an Epstein zeta function instead of citing the Chowla–Selberg formula (Theorem 16.6), and checks it by the route of this section (Remark 16.7).]

7 All orders: existence, remainders, and coefficients

7.1 The convergent local expansion

Write $V(s) = \sum_{j \geq 0} v_j s^j$, $v_0 = 1$, $v_{-1} = 0$. Substitution of $s^{1/2}V(s)$ into (18) gives

$$\begin{aligned} 28j(j + \tfrac{1}{2})v_j &= \left(29(j - \tfrac{1}{2})^2 + 14(j - \tfrac{1}{2}) + \tfrac{20}{9}\right)v_{j-1} \\ &\quad - \left((j - \tfrac{3}{2})^2 + (j - \tfrac{3}{2}) + \tfrac{2}{9}\right)v_{j-2}, \quad j \geq 1. \end{aligned} \quad (69)$$

Thus all v_j are rational, with $v_1 = 593/1512$. Frobenius theory makes this a *convergent local* power series, not merely a formal solution at infinity.

Manuscript 56 writes the same recurrence with integer coefficients,

$$504(m+1)(2m+3)v_{m+1} - (1044m^2 + 1548m + 593)v_m + (36m^2 - 1)v_{m-1} = 0, \quad (70)$$

and lists

$$\begin{aligned} v_1 &= \frac{593}{1512}, & v_2 &= \frac{7493}{31104}, & v_3 &= \frac{400316059}{2304433152}, \\ v_4 &= \frac{5678514224315}{41811635109888}, & v_5 &= \frac{25841240646919967}{231803705049219072}. \end{aligned} \quad (71)$$

(The two recurrences give the same v_j ; the intake checked $j \leq 9$ exactly.) Thus a more detailed version of (68) is

$$A(z) = U(1-27z) - \frac{2\pi\sqrt{3}}{\mathcal{G}_7}(1-27z)^{1/2} \left(1 + \frac{593}{1512}(1-27z) + \frac{7493}{31104}(1-27z)^2 + \dots \right). \quad (72)$$

7.2 Transfer to coefficients

For nonintegral α , the exact coefficient formula is

$$[z^n](1-27z)^\alpha = 27^n \frac{\Gamma(n-\alpha)}{\Gamma(-\alpha)\Gamma(n+1)}. \quad (73)$$

Apply it to $B \sum_{j=0}^M v_j (1-27z)^{j+1/2}$ and expand each gamma ratio [13, §5.11]. Singularity analysis in the dented disk from Section 3 transfers the remainder. The locally analytic part contributes no algebraic terms from this singularity. Equivalently one may subtract arbitrarily many Taylor terms of U before applying the transfer theorem. The first omitted nonintegral power is $(1-27z)^{M+3/2}$, and its coefficient is $O(27^n n^{-M-5/2})$. This proves (4) for every fixed M .

[Merge note, batch 7302: Manuscript 56's version of this step reads: "The gamma quotient has a complete expansion in descending powers of n . Combining the Frobenius series and (73) therefore gives the expansion. The analytic branch contributes only an exponentially smaller term after the local singular part is removed, because the next finite singularity is at $z = -1$." It does not state the continuation domain. The hypothesis singularity analysis needs, continuation to a dented disk of radius greater than ρ with no other singularity on $|z| = \rho$, is manuscript 58's (Section 3); this report relies on it for both proofs.]

The first correction is transparent: the gamma ratio for $\alpha = 1/2$ contributes $3/(8n)$; the $s^{3/2}$ term contributes $-3v_1/(2n)$, relative to the leading term. Hence

$$c_1 = \frac{3}{8} - \frac{3}{2} \frac{593}{1512} = -\frac{215}{1008}.$$

No numerical fitting of the coefficients is needed.

7.3 A triangular recurrence at infinity

For efficient computation insert $u_n = r^n n^{-3/2} S(1/n)$ in the recurrence, with $r \in \{27, -1\}$. The corresponding formal operator is

$$\begin{aligned} \mathcal{E}_r[S](t) &= r(1+t)^{1/2} S\left(\frac{t}{1+t}\right) - (26 + 13t + 2t^2)S(t) \\ &\quad - \frac{27 - 27t + 6t^2}{r} (1-t)^{-3/2} S\left(\frac{t}{1-t}\right). \end{aligned} \quad (74)$$

The characteristic equation is $r^2 - 26r - 27 = 0$, with roots $27, -1$.

Proposition 7.1 (Exact coefficient algorithm). *For either root there is a unique formal series $S_r(t) = 1 + \sum_{j \geq 1} \gamma_j t^j$ with $\mathcal{E}_r[S_r] = 0$. Writing $S_{m-1} = \sum_{j=0}^{m-1} \gamma_j t^j$, one has*

$$\gamma_m = \frac{[t^{m+1}]\mathcal{E}_r[S_{m-1}](t)}{m(r + 27/r)}, \quad m \geq 1. \quad (75)$$

All coefficients are rational; the denominators in the recursion are $28m$ and $-28m$, respectively.

Proof. A newly inserted $\gamma_m t^m$ contributes zero in order m , by the characteristic equation. Its coefficient in order $m + 1$ is

$$r\left(\frac{1}{2} - m\right) - 13 - \frac{27}{r}\left(\frac{1}{2} + m\right) = -m(r + 27/r).$$

This is nonzero, giving the assertion inductively. At the first two orders the normalization is compatible with both characteristic roots. $\square$

For $r = 27$, the analytic existence proof above identifies this formal series with the asymptotic series of a_n/C . For $r = -1$, Section 9 will construct a genuine solution realizing the series. Formal coefficient generation alone is not used to assert existence of an asymptotic expansion.

Manuscript 56 derives the case $r = 27$ independently: with $S_{27} = \mathcal{A}$ in its notation, $\mathcal{E}_{27}[S] = 0$ is its formal identity, since $(27 - 27t + 6t^2)/27 = 1 - t + \frac{2}{9}t^2$ (Appendix B). It records the next five coefficients beyond (5):

$$\begin{aligned} c_6 &= \frac{5073267927236405}{3082688233381822464}, \\ c_7 &= \frac{141980078251330365605}{21751448174742139305984}, \\ c_8 &= \frac{1180675715260847223544061}{175403678081120611363454976}, \\ c_9 &= -\frac{541114776511779121907204365}{227323166793132312327037648896}, \\ c_{10} &= -\frac{10501970724197421668190580597385}{458283504254954741651307900174336}. \end{aligned} \quad (76)$$

These agree with manuscript 58's recorded coefficients, which continue to c_{12} in `data/58-gamma-verification.json`. The existence of the local convergent Frobenius series, together with standard singularity analysis, upgrades the formal computation to the error statement in Theorem 2.1 for every fixed order.

7.4 Logarithms, ratios, and convexity

Let $\log S_{27}(t) = \sum_{j \geq 1} \ell_j t^j$. Then

$$\log a_n = n \log 27 - \frac{3}{2} \log n + \log C + \sum_{j=1}^M \frac{\ell_j}{n^j} + O_M(n^{-M-1}), \quad (77)$$

with

$$\ell_1 = -\frac{215}{1008}, \quad \ell_2 = -\frac{85}{3136}, \quad \ell_3 = \frac{99905}{85349376}, \quad \ell_4 = \frac{1657}{2458624}. \quad (78)$$

In particular,

$$\frac{a_{n+1}}{a_n} = 27 \left(1 - \frac{3}{2n} + \frac{2105}{1008n^2} + O(n^{-3}) \right). \quad (79)$$

(Corollary 2.3, from manuscript 56, adds the terms in n^{-3} and n^{-4} .) The sequence is strictly increasing, since the coefficient of a_n in (1) already exceeds 1 for $n \geq 1$, and $a_1 > a_0$. Taking sufficiently many terms before forming second differences gives

$$\log a_{n+1} - 2 \log a_n + \log a_{n-1} = \frac{3}{2n^2} + O(n^{-3}) > 0$$

for sufficiently large n . Thus it is eventually strictly log-convex. No explicit smallest threshold is asserted.

8 A continued fraction and exact rational bracketing

8.1 The discrete Wronskian

Let b_n be as in Theorem 2.6 and set $W_n = a_n b_{n+1} - a_{n+1} b_n$. Eliminating the two forward terms gives

$$W_n = -\frac{3(3n-1)(3n-2)}{(n+1)^2} W_{n-1}, \quad W_0 = 1.$$

Grouping factors by residue modulo 3 yields the exact formula

$$W_n = (-1)^n \frac{(3n)!}{n!(n+1)!^2}. \quad (80)$$

Consequently

$$\delta_n := \frac{b_{n+1}}{a_{n+1}} - \frac{b_n}{a_n} = (-1)^n \frac{(3n)!}{n!(n+1)!^2 a_n a_{n+1}}. \quad (81)$$

Its magnitude decreases strictly because

$$\left| \frac{\delta_{n+1}}{\delta_n} \right| = \frac{3(3n+2)(3n+1)a_n}{(n+2)^2 a_{n+2}} < 1.$$

The numerator is one positive summand in the recurrence for the denominator. The ratio tends to $1/27$ by Theorem 2.1. Thus $\sum \delta_n$ converges absolutely, and its alternating tails prove the existence of L and the brackets (13). Even approximants increase and odd approximants decrease; consecutive ones enclose L with the exact width

$$\left| \frac{b_{n+1}}{a_{n+1}} - \frac{b_n}{a_n} \right| = \frac{(3n)!}{n!(n+1)!^2 a_n a_{n+1}}. \quad (82)$$

No unknown constants enter this stopping criterion.

8.2 Polynomial continued-fraction coefficients

The rescaled sequences $\mathcal{H}_n = (n!)^2 a_n$ and $\mathcal{K}_n = (n!)^2 b_n$ satisfy

$$u_{n+1} = P_n u_n + Q_n u_{n-1}, \quad P_n = 26n^2 + 13n + 2, \quad Q_n = 3n^2(3n-1)(3n-2), \quad (83)$$

with initial pairs $(1, 2)$ and $(0, 1)$. Their integrality follows directly from this recurrence, independently of integrality of the unscaled sequence. The continuant recurrence gives

$$L = \frac{1}{2 + \frac{6}{41 + \frac{240}{132 + \frac{1512}{275 + \ddots}}}}. \quad (84)$$

The convergent with n denominator levels is $\mathcal{K}_n/\mathcal{H}_n = b_n/a_n$.

Remark 8.1 (The initial relation matters). The companion satisfies the recurrence only for $n \geq 1$; $b_1 = 2b_0$ is false. Its generating function $\mathcal{B}(z) = \sum b_n z^n$ therefore satisfies (17) with right side 1, not 0. The same holds for $\mathcal{B} - LA$. These must not be called second analytic solutions of the homogeneous equation at 0. The two-dimensional solution space here consists of sequences satisfying the recurrence for $n \geq 1$.

9 The recessive scale and its full expansion

Stirling's formula in (80) gives

$$W_n = (-1)^n \frac{\sqrt{3}}{2\pi} \frac{27^n}{n^3} (1 + O(n^{-1})). \quad (85)$$

Together with the dominant asymptotic this yields

$$\delta_n = (-1)^n \frac{\sqrt{3}}{54\pi C^2} 27^{-n} (1 + O(n^{-1})). \quad (86)$$

The exact tail identities are

$$L - \frac{b_n}{a_n} = \sum_{j \geq 0} \delta_{n+j}, \quad e_n = -a_n \sum_{j \geq 0} \delta_{n+j}. \quad (87)$$

Since $\sum_{j \geq 0} (-1/27)^j = 27/28$, the leading terms of (16) and (14) follow. The recessive amplitude can also be written

$$D_{\min} = \frac{\sqrt{3}}{56\pi C} = \frac{\mathcal{G}_7}{56\pi^{3/2}}. \quad (88)$$

9.1 A remainder proof for the geometric tail

The full Stirling expansion and Theorem 2.1 imply, for each fixed M ,

$$\delta_n = (-1)^n K_0 27^{-n} \left(\sum_{k=0}^M \eta_k n^{-k} + O_M(n^{-M-1}) \right), \quad K_0 = \frac{\sqrt{3}}{54\pi C^2}, \quad \eta_0 = 1,$$

where all η_k are rational. The error bound is valid for all sufficiently large integer arguments, hence also at $n + j$ uniformly in $j \geq 0$, with a bound by a constant times $27^{-j} n^{-M-1}$ after factoring out 27^{-n} . Expand $(n + j)^{-k}$ for $0 \leq j \leq n/2$ by Taylor's formula, retaining its bounded remainder. The range $j > n/2$ is exponentially small, even with any fixed polynomial weight. Every coefficient involves moments

$$\sum_{j \geq 0} j^h (-1/27)^j,$$

which are rational: apply $(q d/dq)^h$ to $1/(1-q)$ and set $q = -1/27$. Thus the tail in (87) has an expansion to every fixed order with a controlled error and rational normalized coefficients. Multiplication by the expansion of a_n proves existence of (14). Since e_n satisfies the recurrence, Proposition 7.1 with $r = -1$ proves that its coefficients are precisely the d_j already given. This completes the proof of Theorem 2.6.

9.2 What the two-scale expansion does and does not say

Because $W_n \neq 0$, a_n and e_n form an exact basis of the recurrence solutions for $n \geq 1$. Every solution has a unique form $u_n = \alpha a_n + \beta e_n$. In this specified basis its two asymptotic scales are

$$u_n \sim \alpha C 27^n n^{-3/2} \sum_{j \geq 0} c_j n^{-j} + \beta (-1)^{n+1} D_{\min} n^{-3/2} \sum_{j \geq 0} d_j n^{-j}. \quad (89)$$

Each exact basis sequence has its own fixed-order expansion. A finite truncation of the dominant expansion generally has an error far larger than the entire recessive term. Equation (89) is therefore not a claim that two finite sums resolve exponentially small effects in a generic solution. A canonical exponentially improved expansion of a_n alone would require additional summation or Stokes data; that question is left open.

10 Convolution powers and A183204

For a fixed positive integer m , let

$$a_n^{(m)} = [z^n] A(z)^m = \sum_{n_1 + \dots + n_m = n} a_{n_1} \cdots a_{n_m}.$$

The exact boundary value controls this family as well as the original sequence.

Theorem 10.1 (Convolution asymptotics; manuscript 58). *For every fixed $m \geq 1$,*

$$a_n^{(m)} = m A_\star^{m-1} C \frac{27^n}{n^{3/2}} \left(1 + \frac{c_1^{(m)}}{n} + O_m(n^{-2}) \right), \quad (90)$$

where

$$c_1^{(m)} = -\frac{215}{1008} - \frac{5}{21}(m-1) - \pi \frac{C^2}{A_\star^2} (m-1)(m-2). \quad (91)$$

There is a computable expansion to every fixed order, and all its normalized coefficients lie in $\mathbb{Q}(\pi C^2/A_\star^2)$.

Proof. The analytic part of (19) satisfies

$$U(s) = A_\star \left(1 + \frac{10}{63}s + O(s^2) \right),$$

as the constant coefficient of (18) shows. Its subsequent normalized coefficients are rational. The singular part is $B\sqrt{s}(1 + v_1 s + \dots)$, with $B = -2\sqrt{\pi}C$ and $v_1 = 593/1512$. In the m th power the coefficient of $\sqrt{s}$ is $m A_\star^{m-1} B$. Relative to that coefficient, the coefficient of $s^{3/2}$ is

$$v_1 + (m-1) \frac{10}{63} + \frac{(m-1)(m-2)}{6} \frac{B^2}{A_\star^2}.$$

Transfer as in Section 7 gives (91). At higher orders only odd powers of B multiply half-integral powers of s . After division by the leading singular coefficient, the remaining expressions are polynomials in $B^2/A_\star^2 = 4\pi C^2/A_\star^2$ with rational coefficients. The convergent Puiseux expansion and the same dented domain provide the fixed-order remainder estimates. $\square$

Remark 10.2 (An explicit generator for every order; manuscript 32). [Write note, 2026-10-02, batch 77P5: Manuscript 32 proves Theorem 10.1 again (its Theorem 7.1, with weight $k = m$) and makes the “computable expansion to every fixed order” explicit. Notation is that of Table 7.] Put

$$\mathfrak{q} = \frac{B^2}{A_\star^2} = \frac{4\pi C^2}{A_\star^2} = \frac{256\pi^6}{3\mathcal{G}_7^4} = 0.0768944281311685853765\dots,$$

the quantity $B^2/A_\star^2$ of the proof above (it is *not* the hypergeometric argument $q_0 = 64/85^3$ of Section 5; manuscript 32’s replay output calls it $\mathfrak{q}\emptyset$). With $U(s) = A_\star \widehat{U}(s)$, $\widehat{U}(0) = V(0) = 1$, the coefficient of $s^{j+1/2}$ in $A(z)^m = (A_\star \widehat{U}(s) + B\sqrt{s} V(s))^m$ is $B_j^{(m)} = mA_\star^{m-1} B \mathcal{R}_{m,j}$, where the binomial theorem gives the finite sum

$$\mathcal{R}_{m,j} = \sum_{h=0}^{\min(j, \lfloor (m-1)/2 \rfloor)} \frac{1}{m} \binom{m}{2h+1} \mathfrak{q}^h [s^{j-h}] \widehat{U}(s)^{m-2h-1} V(s)^{2h+1}, \quad \mathcal{R}_{m,0} = 1.$$

Transfer as in Section 7.2 then gives, for every fixed M ,

$$a_n^{(m)} = 27^n \sum_{j=0}^M B_j^{(m)} \frac{\Gamma(n-j-\frac{1}{2})}{\Gamma(-j-\frac{1}{2})\Gamma(n+1)} + O_{m,M}(27^n n^{-M-5/2}),$$

and, with the gamma-ratio coefficients h_i of (16.14),

$$c_i^{(m)} = \sum_{j=0}^i \mathcal{R}_{m,j} (-1)^j \left(\frac{3}{2}\right)_j h_{i-j}(-j-\frac{1}{2}, 1).$$

All operations are finite and all exponents nonnegative integers, so $c_i^{(m)}$ lies in the polynomial ring $\mathbb{Q}[\mathfrak{q}]$, a sharper form of the field statement of Theorem 10.1. For $i = 1$, $\mathcal{R}_{m,1} = \frac{593}{1512} + \frac{10}{63}(m-1) + \frac{(m-1)(m-2)}{6}\mathfrak{q}$ and $h_1(-\frac{1}{2}, 1) = \frac{3}{8}$, so $c_1^{(m)} = \frac{3}{8} - \frac{3}{2}\mathcal{R}_{m,1}$, which is (91) because $\pi C^2/A_\star^2 = \mathfrak{q}/4 = 64\pi^6/(3\mathcal{G}_7^4)$; for $m = 3$ it is $-0.72793134104971127682\dots$. For $m \leq 2$ only $h = 0$ occurs, so every $c_i^{(1)}$ and $c_i^{(2)}$ is rational; for $m \geq 3$ the first correction is a polynomial of degree one in $\mathfrak{q}$. For $m = 2$ the leading constant and $c_1^{(2)} = -65/144$ are Hirschhorn’s (92); manuscript 32 recovers them as a consistency check on prior work, not as a new result, and it checks the binomial sum (37) only over a finite range (its replay: $n \leq 40$). It asserts no OEIS identification for $m \geq 3$.

Manuscript 32 also records that each $a_n^{(m)}$ is strictly increasing in n : raising the index of the first factor maps the terms of degree n injectively to strictly larger terms of degree $n+1$, and the remaining terms of degree $n+1$ are positive. Its replay checks the local expansion of A^m by direct series multiplication for $m = 1, \dots, 6$ and the convolutions for $m = 2, 3$ through $n = 400$. These are finite checks.

For $m = 2$ this is A183204, and the first two terms simplify to

$$[z^n]A(z)^2 = \frac{3\sqrt{3}}{4\pi^{3/2}} \frac{27^n}{n^{3/2}} \left(1 - \frac{65}{144n} + O(n^{-2})\right). \quad (92)$$

The leading formula is already recorded in its OEIS entry [2]; it is recovered here, not claimed as new. The cancellation of gamma factors at this convolution power is a useful check on the evaluation of C . (Both terms are Hirschhorn’s [4], as quoted by manuscript 56 in (48), which runs on to the n^{-3} term. In the second proof of Section 6, the logic is reversed: the leading constant of (92) is the input from which C is extracted.)

For an algebraic check of the generating-function identification, the symmetric-square equation of (17) is

$$\begin{aligned} &(-z^2 + 26z^3 + 27z^4)Y''' + (-3z + 117z^2 + 162z^3)Y'' \\ &\quad + (-1 + 86z + 186z^2)Y' + (4 + 24z)Y = 0. \end{aligned} \quad (93)$$

This is the equation recorded for A183204, and its normalized analytic solution is unique. The program verifies this symmetric-square identity and checks the convolution against

$$\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{n} \binom{2n-k}{n}$$

for $0 \leq n \leq 40$. The finite check supplements the differential-equation argument; it does not replace it.

11 Inversion and the integer-rounding boundary

[Merge note, batch 73O2: Both manuscripts invert the asymptotic by Lambert's W_{-1} . The core and its triangular corrections are the instance $a = \lambda = \log 27$, $\beta_{\text{CTI}} = -3/2$, $q_j^{\text{CTI}} = \ell_j$ of Theorem `t2:thm:balanced-inverse` of the repository's volume *Combinatorial Transseries and Their Inverses* [16]: the coefficients q_1, q_2 of (97) below are that theorem's d_1, d_2 (checked by hand at intake). Neither manuscript cites it, and this report claims no novelty for the inverse expansion. The rounding proposition, Proposition 11.2, is the sequence-specific part; its conclusion has the form of the separation condition (part 2) of Theorem `p0:thm:staircase` in the repository's canonical volume *Transseries and Inversion* [17].]

Put $\lambda = \log 27$ and $\beta = 3/2$. The leading smooth model is $Y = Ce^{\lambda x} x^{-\beta}$. Its large positive solution is

$$x_0(Y) = -\frac{\beta}{\lambda} W_{-1} \left(-\frac{\lambda}{\beta} (C/Y)^{1/\beta} \right), \quad (94)$$

where W_{-1} is the real negative branch of Lambert's function. The principal branch selects the small solution and is inappropriate for inversion of the large- n sequence [13, §4.13]. (Manuscript 56 derives the same formula, raising the envelope to the power $-2/3$.)

11.1 A triangular inverse expansion

Seek

$$x_M(Y) = x_0 + \sum_{j=1}^M \frac{q_j}{x_0^j}. \quad (95)$$

Writing $\delta = x - x_0$, the formal equation determining the coefficients is

$$\lambda \delta - \beta \log(1 + \delta/x_0) + \sum_{k \geq 1} \ell_k (x_0 + \delta)^{-k} = 0. \quad (96)$$

The newly introduced coefficient q_j has multiplier λ , so this is triangular at every order. The first coefficients are

$$\begin{aligned} q_1 &= -\frac{\ell_1}{\lambda} = \frac{215}{1008\lambda}, \\ q_2 &= \frac{\beta q_1 - \ell_2}{\lambda} = \frac{215}{672\lambda^2} + \frac{85}{3136\lambda}, \\ q_3 &= \frac{\beta q_2 + \ell_1 q_1 - \ell_3}{\lambda}. \end{aligned} \quad (97)$$

All q_j belong to $\mathbb{Q}(\lambda)$.

An expansion involving only logarithms follows by putting $X = \log(Y/C)$:

$$x(Y) = \frac{1}{\lambda} \left\{ X + \beta \log \frac{X}{\lambda} + \frac{\beta^2 \log(X/\lambda) - \lambda \ell_1}{X} \right\} + O\left(\frac{(\log X)^2}{X^2}\right). \quad (98)$$

Here $x(Y)$ denotes the asymptotic inverse scale, not a previously specified analytic interpolation of the integer sequence. Formula (94) is often more accurate numerically because it retains the entire leading model instead of truncating its logarithmic inversion.

Remark 11.1 (Manuscript 56's form of the inverse). Expanding W_{-1} at 0^- , manuscript 56 records

$$x_0(Y) = \frac{1}{\lambda} \left(\log Y + \frac{3}{2} \log \log Y - \frac{3}{2} \log \lambda - \log C \right) + O\left(\frac{\log \log Y}{\log Y}\right), \quad (99)$$

consistent with the first two terms of (98). It notes that the rational correction series can be inserted and reverted recursively to produce an arbitrarily long expansion in $1/\log Y$ with nested powers of $\log \log Y$, and that for the discrete inverse $N(Y)$ below, formula (94), “rounded and corrected by one or two exact recurrence evaluations, gives an efficient certified locator”. [*Merge note, batch 73O2*: The last statement is a practical recipe, not a theorem: a *certified* locator needs explicit constants K_M, Y_M in Proposition 11.2, which neither manuscript supplies (manuscript 58 lists them as a research direction, Section 13.1). Checking a candidate n by exact evaluation of $a_{n-1} < Y \leq a_n$ is of course rigorous.]

11.2 A rigorous discrete statement

Define

$$N(Y) = \min\{n \geq 0 : a_n \geq Y\}.$$

The sequence is strictly increasing and unbounded, so this is well-defined for $Y > 1$. An inverse approximation cannot uniformly remove the ambiguity when its value is exceptionally close to an integer.

Proposition 11.2 (Rounding brackets). *For every fixed $M \geq 0$ there exist $K_M > 0$ and Y_M such that, for $Y \geq Y_M$,*

$$\lceil x_M - K_M x_M^{-M-1} \rceil \leq N(Y) \leq \lceil x_M + K_M x_M^{-M-1} \rceil. \quad (100)$$

When the two ceilings coincide, they determine the discrete inverse exactly. For $M = 0$, x_M means x_0 .

Proof. Use the smooth finite model

$$g_M(x) = \log C + \lambda x - \beta \log x + \sum_{k=1}^M \ell_k x^{-k}.$$

The all-orders theorem gives $\log a_n = g_M(n) + O_M(n^{-M-1})$. Also $g'_M(x) = \lambda + O(x^{-1})$, so it is bounded away from zero for large x . Let t_M be the large solution of $g_M(t_M) = \log Y$. Formal substitution of (95) into g_M , with ordinary finite Taylor remainders, gives $g_M(x_M) - \log Y = O_M(x_M^{-M-1})$. The mean value theorem therefore implies $t_M - x_M = O_M(x_M^{-M-1})$.

For integers near t_M , comparison of $g_M(n)$ with $g_M(t_M)$, using its derivative lower bound and the error in $\log a_n$, shows that integers sufficiently below t_M cannot reach Y , whereas integers sufficiently above it do reach Y . Enlarge the constant to absorb $t_M - x_M$, and use monotonicity of a_n for all remaining integers. This gives the two ceilings in (100). $\square$

The proposition asserts the existence of fixed-order constants, not explicit numerical values of K_M or Y_M . Computing certified, practical values is one of the research directions below. No claim that $N(Y) = \lceil x_M(Y) \rceil$ holds uniformly is made.

Remark 11.3 (Every fixed weight, and an implicit-function proof; manuscript 32). [Write note, 2026-10-02, batch 77P5: Manuscript 32 states the rounding brackets for every fixed convolution weight m (its Theorems 8.1 and 8.2, weight $k = m$), with notation as in Table 7. For $m = 1$ they are Proposition 11.2. The Lambert- W_{-1} core and its triangular corrections are, as for $m = 1$, instances of the repository's inversion apparatus: `t2:thm:balanced-inverse` of [16], and `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of [17]. Neither manuscript 32 nor this report claims novelty for them.] Let $N^{(m)}(Y) = \min\{n \geq 0 : a_n^{(m)} \geq Y\}$, well defined for every real $Y > 0$ because $a_n^{(m)}$ is strictly increasing and unbounded (Remark 10.2). With $C^{(m)} = mA_{\star}^{m-1}C$ and $S_M^{(m)}(u) = \sum_{j=0}^M c_j^{(m)} u^j$, define the smooth model

$$\mathcal{F}_M^{(m)}(v) = C^{(m)} e^{\lambda v} v^{-\beta} S_M^{(m)}(1/v), \quad \frac{d}{dv} \log \mathcal{F}_M^{(m)}(v) = \lambda + O_{m,M}(1/v),$$

and let $\tilde{x}_M^{(m)}(Y)$ be its unique large solution of $\mathcal{F}_M^{(m)}(v) = Y$. Then there are $K_M^{(m)} > 0$ and $Y_M^{(m)}$ such that, for $Y \geq Y_M^{(m)}$ and $\tilde{x} = \tilde{x}_M^{(m)}(Y)$,

$$\lceil \tilde{x} - K_M^{(m)} \tilde{x}^{-M-1} \rceil \leq N^{(m)}(Y) \leq \lceil \tilde{x} + K_M^{(m)} \tilde{x}^{-M-1} \rceil.$$

The proof is that of Proposition 11.2: the forward expansion gives $|\log a_n^{(m)} - \log \mathcal{F}_M^{(m)}(n)| \leq Dn^{-M-1}$; the logarithmic derivative is at least $\lambda/2$ beyond a fixed point; for $\epsilon = K\tilde{x}^{-M-1}$ the mean value theorem gives $a_{n_+}^{(m)} \geq Y$ at $n_+ = \lceil \tilde{x} + \epsilon \rceil$ and $a_{n_-}^{(m)} < Y$ at $n_- = \lceil \tilde{x} - \epsilon \rceil - 1$, and monotonicity gives $n_- + 1 \leq N^{(m)}(Y) \leq n_+$.

For the centre, let $x_0^{(m)}(Y)$ be (94) with C replaced by $C^{(m)}$, let $\ell_j^{(m)} = [u^j] \log \sum_{i \geq 0} c_i^{(m)} u^i$ (a formal logarithm; convergence of the coefficient series is not assumed), and define $q_j^{(m)}$ by $\delta(v) = \sum_{j \geq 1} q_j^{(m)} v^j$ and

$$\lambda \delta - \beta \log(1 + v\delta) + \sum_{j \geq 1} \ell_j^{(m)} \left(\frac{v}{1 + v\delta} \right)^j = 0,$$

triangular because the new $q_j^{(m)}$ enters with multiplier λ . Then $\tilde{x}_M^{(m)}(Y) = x_0^{(m)} + \sum_{j=1}^M q_j^{(m)} (x_0^{(m)})^{-j} + O_{m,M}((x_0^{(m)})^{-M-1})$. Manuscript 32 proves this by the *analytic* implicit-function theorem rather than by formal substitution: with $\tilde{x} = x_0^{(m)} + \delta$ and $v = 1/x_0^{(m)}$, the exact equation $\lambda \delta - \beta \log(1 + v\delta) + \log S_M^{(m)}(v/(1 + v\delta)) = 0$ is analytic near $(0, 0)$ with δ -derivative $\lambda \neq 0$, so it has a convergent solution $\delta(v)$, which is the large inverse for small $v > 0$; its first M Taylor coefficients are the $q_j^{(m)}$, and raising M leaves the determined coefficients unchanged. Explicitly $q_1^{(m)} = -\ell_1^{(m)}/\lambda$, $q_2^{(m)} = -\beta \ell_1^{(m)}/\lambda^2 - \ell_2^{(m)}/\lambda$ and $q_3^{(m)} = -\beta^2 \ell_1^{(m)}/\lambda^3 - (\beta \ell_2^{(m)} + (\ell_1^{(m)})^2)/\lambda^2 - \ell_3^{(m)}/\lambda$; for $m = 1$ these are q_1, q_2, q_3 of (97) (checked by hand at this write). The replay records the universal coefficients through order 6. Combining the two statements, the centre $\tilde{x}$ of the brackets may be replaced by the truncated expansion after enlarging the constant; for $m = 1$ this is Proposition 11.2 with centre x_M .

As in Proposition 11.2, $K_M^{(m)}$ and $Y_M^{(m)}$ are existential. Once evaluated effectively, the brackets decide $N^{(m)}(Y)$ whenever the two ceilings agree; without such values they supply no practical rounding certificate, and no unqualified exact ceiling at or near a threshold is asserted. The replay's inverse residuals are taken at $Y = a_n^{(m)}$, exactly at discrete thresholds, and are not a ceiling rule.

12 Reproducible computations and certified constants

[Merge note, batch 7302: File names in this section are the shipped ones. Manuscript 58 named its README, SOURCE_AUDIT.md and data/verification.json; they are this report's README.md, 58-gamma-SOURCE_AUDIT.md and data/58-gamma-verification.json.]

Build and run instructions are in the accompanying README.md; manuscript 58's source audit is in 58-gamma-SOURCE_AUDIT.md. The PDF can be built without running Python, since its numerical tables are part of the TeX source. Manuscript 58's program (code/58-gamma-verify.py) uses exact integers and rational functions for its algebraic checks. Arbitrary-precision floating-point arithmetic is used separately for numerical diagnostics. It requires only Python, SymPy, and mpmath, and makes no network requests.

12.1 Certificates and reproducible output

The following assertions were executed successfully: eight rational identities for the two pullbacks and two symmetric squares; coefficient recurrences through order 12 for both characteristic roots; exact recurrence evaluation through $n = 1000$; 100 discrete Wronskian identities; and 41 comparisons with the binomial sum for A183204. The division in the a_n recurrence is exact at every tested index. This last computation is a finite integrality check, not an independent proof of integrality for all indices.

The constants are

$$\begin{aligned} C &= 0.09552230526812671465130791078702962567279466665100\dots, \\ A_\star &= 1.22113109746756666928422458568599346237582705637935\dots, \\ L &= 0.46721176888437312394211005141309126172410854995971\dots, \\ D_{\min} &= 0.10306659600794607491874204424062776206902297866964\dots \end{aligned}$$

The first displayed value is not inferred by integer-relation searching: its gamma formula is proved in Section 5 (and again in Section 6), and the following interval method supplies independent certified decimal evaluation.

12.2 A rational enclosure for the asymptotic constant

Write $F(q_0) = \sum_{j \geq 0} h_j$. Its positive terms satisfy

$$\frac{h_{j+1}}{h_j} = q_0 \frac{(12j+1)(12j+5)}{144(j+1)^2} < q_0 < 1.$$

Hence, for $S_N = \sum_{j=0}^{N-1} h_j$,

$$S_N \leq F(q_0) \leq S_N + \frac{h_N}{1 - q_0}. \quad (101)$$

Every quantity in this enclosure is rational. The implementation bounds π using Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and exact alternating-series bounds. Square and fourth roots are bracketed by rational numbers using integer root extraction on a fixed decimal grid. Combining positive upper and lower bounds in (32) gives a rational enclosure for C of width less than 10^{-120} . Outward rounding produces the following interval of width 10^{-100} ; each row continues on its second line.

Lower: 0.09552230526812671465130791078702962567279466665100
 71798669948234917659264524531457675812093972181446
 Upper: 0.09552230526812671465130791078702962567279466665100
 71798669948234917659264524531457675812093972181447

For L the enclosure uses no transcendental functions at all: $b_{90}/a_{90} < L < b_{91}/a_{91}$, with exact rational endpoints whose difference is less than 10^{-125} . The full endpoints and their 100-place decimal enclosures are stored in `data/58-gamma-verification.json`. The JSON also records every coefficient through order 12 and the zero values of all eight rational certificates.

12.3 Numerical diagnostics

For Table 2, the relative error is

$$\frac{C27^n n^{-3/2} \sum_{j=0}^M c_j n^{-j}}{a_n} - 1.$$

The values are rounded floating-point diagnostics, not new rigorous upper bounds on the asymptotic remainder. All were computed with 190 decimal digits of working precision.

Table 2: Dominant expansion: signed relative errors (manuscript 58).

n	$M = 0$	$M = 3$	$M = 6$	$M = 12$
20	1.079×10^{-2}	-1.143×10^{-9}	-5.413×10^{-12}	2.818×10^{-19}
100	2.138×10^{-3}	-2.449×10^{-12}	-6.609×10^{-17}	6.083×10^{-28}
1000	2.133×10^{-4}	-2.597×10^{-16}	-6.536×10^{-24}	6.727×10^{-41}

The normalized recessive error in Table 3 is $\sum_{j=0}^M d_j n^{-j} - e_n / [(-1)^{n+1} D_{\min} n^{-3/2}]$. The value of L used in this diagnostic is the midpoint of the exact r_{150}, r_{151} enclosure, where $r_n = b_n/a_n$. Its error remains negligible after multiplication by a_n at the displayed indices.

Table 3: Recessive expansion: signed normalized errors (manuscript 58).

n	$M = 0$	$M = 3$	$M = 6$
10	4.910×10^{-2}	1.721×10^{-6}	-1.532×10^{-9}
30	1.676×10^{-2}	2.361×10^{-8}	-6.532×10^{-13}
75	6.753×10^{-3}	6.240×10^{-10}	-1.036×10^{-15}

Table 4 tests inversion at $Y = a_n$. It reports the signed continuous error $x_M(a_n) - n$, not an integer rounding error. The uncorrected Lambert- W scale is already accurate; the algebraic corrections produce the predicted further improvement.

Table 4: Inverse expansion at exact sequence values (manuscript 58).

n	$x_0(a_n) - n$	$x_1(a_n) - n$	$x_3(a_n) - n$
20	-3.332×10^{-3}	-9.581×10^{-5}	-6.315×10^{-9}
100	-6.509×10^{-4}	-3.780×10^{-6}	-1.064×10^{-11}
1000	-6.475×10^{-5}	-3.769×10^{-8}	-1.075×10^{-15}

12.4 Manuscript 56's tables and checks

All numerical checks of manuscript 56 were generated from exact integers using (1); no OEIS term table was used after the initial comparison. Define the scaled quantity

$$R_n = \frac{a_n n^{3/2}}{27^n}. \quad (102)$$

Table 5 compares R_n with the exact constant. The last column subtracts the approximation through c_3/n^3 and divides the residual by C .

Table 5: Convergence to the exact asymptotic constant (manuscript 56).

n	R_n	$R_n/C - 1$	residual after c_3
10	0.0934812234164389792	$-2.1367594 \times 10^{-2}$	1.1427×10^{-8}
50	0.0951146568282489133	$-4.2675733 \times 10^{-3}$	3.6462×10^{-11}
100	0.0953185211418410836	$-2.1333669 \times 10^{-3}$	2.4441×10^{-12}
500	0.0954815550048036949	$-4.2660469 \times 10^{-4}$	4.1263×10^{-15}
1000	0.0955019305511748892	$-2.1329800 \times 10^{-4}$	2.5960×10^{-16}
5000	0.0955182303912366763	$-4.2658904 \times 10^{-5}$	4.1754×10^{-19}

The residual column is the $M = 3$ column of Table 2 with the opposite sign convention: manuscript 56 tabulates $R_n/C - \sum_{j \leq 3} c_j n^{-j}$, manuscript 58 tabulates $C \sum_{j \leq 3} c_j n^{-j}/R_n - 1$, which is minus the former times $C/R_n = 1 + O(n^{-1})$. At $n = 100$ and 1000 the two agree up to sign to the printed digits (2.4441×10^{-12} against -2.449×10^{-12} , and 2.5960×10^{-16} against -2.597×10^{-16}).

The critical series converges slowly, exactly as Corollary 2.5 predicts. Table 6 gives the exact remainder computed from (62) and its ratio to the leading estimate $2C/\sqrt{N+1}$.

Table 6: Critical-series tail at $z = 1/27$ (manuscript 56).

N	$A(1/27) - \sum_{n=0}^N a_n/27^n$	ratio to $2C/\sqrt{N+1}$
10	0.058541895880356	1.016314999
50	0.026845518682688	1.003510906
100	0.019043336251750	1.001772100
500	0.008538291869020	1.000357123
1000	0.006039421866043	1.000178732
5000	0.002701605297719	1.000035774

Manuscript 56's script (code/56-exact-verify_a279619.py; its recorded output is data/56-exact-verification.txt) performs the following independent checks:

1. generates $a_0, \dots, a_N$ as exact Python integers and verifies divisibility by $(n+1)^2$ at every recurrence step;
2. derives the c_j by exact symbolic expansion of the formal identity $\mathcal{E}_{27}[S] = 0$ of (74);
3. derives the v_j by the exact recurrence (70);
4. evaluates the gamma products and both special sums at high precision;
5. prints Tables 5 and 6;

6. checks the algebraic identities for $A(1/27)$, B , C , and the square-sequence constant C_t .

Items 4 and 6 are high-precision numerical consistency checks of identities proved in Section 6; they are not proofs.

13 Further research questions and topics

The questions below are concrete extensions of the proved results. They are not assertions that every item is globally open: before a priority claim, the relevant modular-form and continued-fraction literature should be checked in greater depth. Sections 13.1–13.12 are manuscript 58’s twelve; Section 13.13 lists manuscript 56’s ten and records where they meet manuscript 58’s.

13.1 Explicit remainder constants

Theorems 2.1 and 2.6 give remainders at every fixed order but not practical values of the constants. Construct computable K_M, N_M such that

$$\left| \frac{a_n n^{3/2}}{C 27^n} - \sum_{j=0}^M c_j n^{-j} \right| \leq K_M n^{-M-1} \quad (n \geq N_M).$$

One route is to bound the analytic Frobenius functions on a specified complex neighborhood and make the transfer contour explicit. A second is to construct upper and lower recurrence solutions from truncated series plus a controlled correction. The latter may give smaller thresholds and could turn the inverse brackets into a practical certified algorithm.

13.2 Late-order growth and optimal truncation

Determine the large- j behavior of c_j and d_j . The proof of fixed-order asymptotics does not imply convergence of the series in $1/n$. A useful target is a sharp factorial-over-power law, with its constant and phase, or a proof of a specified Gevrey bound. This would identify an optimal truncation index and quantify what accuracy is available before the recessive scale becomes relevant.

13.3 Canonical exponentially improved asymptotics

The exact basis a_n, e_n removes ambiguity from (89), but it does not specify a canonical summation of the dominant formal series. Develop a Borel or contour normalization and compute the associated connection data between the singularities ρ and -1 . The desired result is an error theorem resolving the alternating contribution after optimal truncation, not merely a formal addition of a second exponential scale.

13.4 Recognizing the continued-fraction limit

Find an exact special-value expression for L . The inhomogeneous operator in the remark following (84) provides a starting point: variation of parameters expresses its solutions through integrals involving A and algebraic functions. The condition that the $\sqrt{1-27z}$ coefficient vanish selects $\mathcal{B} - LA$ and hence L . A successful reduction might involve an elliptic integral, an iterated modular integral, or a special L -value. No such identification is proved in this report.

13.5 Arithmetic strength of the approximants

Does the continued fraction prove irrationality of L , or a useful irrationality measure? The exponential error 27^{-n} alone is not enough: the natural continuant denominators contain factorial growth. The first step is to analyze the reduced denominators and common factors of $\mathcal{H}_n$ and $\mathcal{K}_n$. Any arithmetic conclusion must compare the error with the actual denominator after cancellation. The general results in [14] are relevant context, but do not automatically apply to the specific limit constructed here.

13.6 The arithmetic conjecture in the entry

The OEIS entry also displays Lucas congruences for primes in the specified residue classes modulo 7 [1]. The present asymptotic argument neither proves nor disproves them. A separate project should first audit subsequent work on the Chan–Cooper–Sica conjecture and then exploit the modular parametrization or the functional equation added to OEIS in 2026 (printed as (63)). This is a different target from the asymptotic conjecture resolved here; analytic evidence should not be substituted for an arithmetic proof.

13.7 Coefficient fields for convolution powers

Theorem 10.1 bounds the coefficient field by $\mathbb{Q}(\pi C^2/A_\star^2)$. Determine exactly which convolution powers and which correction orders eliminate the period ratio. The square is special at its first correction, and its third-order differential equation may expose cancellations more systematically. A symbolic classification would clarify when a convolution inherits purely rational normalized corrections and when new period combinations necessarily appear. [Write note, 2026-10-02, batch 77P5: Partly answered by manuscript 32 (Remark 10.2): every normalized coefficient $c_i^{(m)}$ is an explicit polynomial in $q = B^2/A_\star^2 = 4\pi C^2/A_\star^2$ with rational coefficients; for $m = 1$ and $m = 2$ every correction is rational, at every order; for $m \geq 3$ the first correction is a polynomial of degree one in q , so it is rational only if q is, and the rationality of $q = 256\pi^6/(3\mathcal{G}_7^4)$ is not known. Still open: for $m \geq 3$, whether any higher correction loses its dependence on q , and the arithmetic nature of q .]

13.8 Noninteger powers and growing convolution order

Extend the convolution theorem to a specified branch of $A(z)^u$ for noninteger real or complex u . Zero avoidance and possible additional singularities must be addressed rather than assumed. A separate uniform problem lets the integer $m = m(n)$ grow. The fixed- m proof is then insufficient because powers of the analytic boundary part and the singular part can compete. Determine the applicable uniform ranges and the transition scales from the actual local expansion.

13.9 Positive integral representations

Find a direct integral or moment representation for a_n that makes positivity and possible stronger inequalities transparent. A compact real support reflecting both characteristic roots would be especially informative, but such a representation and positivity of its density are not established here. Potential consequences include explicit log-convexity thresholds, Hankel determinant signs, and alternative error bounds for coefficient asymptotics.

13.10 Automated connection-constant recognition

For families of quadratic-coefficient recurrences with two distinct characteristic roots, combine three exact procedures: derive the Fuchsian equation, certify a rational hypergeometric pullback, and test whether a singular endpoint is a known complex-multiplication value. The intended output should include a symbolic certificate and a branch check, not just a high-precision guess. A279619 is a compact test case because the rational parameter turns the dominant endpoint into $w = 0$.

13.11 A direct modular proof of the product identity

Can the product $CA_\star = 3\sqrt{3}/(8\pi^{3/2})$ be obtained directly from a single level-seven modular relation? The proof here combines a Ramanujan identity with a period evaluation. A direct connection or Wronskian argument on the modular side may unify the two normalizations and generalize to other Apéry-like equations. Such a proof could also explain which sequences exhibit reciprocal boundary and singular periods. [*Merge note, batch 7302*: Manuscript 56’s proof (Section 6) obtains $CA_\star$ as half of Hirschhorn’s constant C_l of the square, and $A_\star$ from a single CM eta product. It is modular on the boundary value but still takes the singular coefficient from the separately proved asymptotic of A183204, so the question as posed, a single modular relation giving the product, remains open.] [*Write note, 2026-10-02, batch 77P5*: Answered in substance for A279619 by manuscript 32 (Proposition 16.2). Differentiating the Fricke involution $z_7(-1/(7\tau)) = -i\sqrt{7}\tau z_7(\tau)$ at its fixed point τ_* gives $BA_\star = -3\sqrt{3}/(4\pi)$, that is $CA_\star = 3\sqrt{3}/(8\pi^{3/2})$; the only other input is O’Brien’s identity $dw_7/d\tau = 2\pi i w_7 z_7^2$, which converts the τ -derivative into a derivative in $\sqrt{1-27z}$. No hypergeometric identity, no period evaluation and no input from A183204 is used. The mechanism is not new: it is the level-seven case of Guillera and Zudilin’s modular limit formula [18, eq. (27)], which is not specific to level seven. What remains open is the last part of the question: which sequences exhibit reciprocal boundary and singular periods, and why.]

13.12 Machine-checked certificates and analytic infrastructure

A staged Lean development can begin with the recurrence, positivity, integer continuants, and the exact Wronskian. These statements need only algebra and induction. The rational identities in (26) and the triangular coefficient algorithm form a second self-contained layer. A later analytic layer must connect convergent local solutions to coefficient asymptotics, including branch choices and remainder estimates. The complex-multiplication identities should be stated as explicit dependencies until they too are formalized. This separation prevents an exact symbolic certificate from being mistaken for a proof of analytic continuation or of a special value.

13.13 Manuscript 56’s questions

Manuscript 56 wrote: the exact constant closes one visible gap, but it exposes several deeper ones. Its list is ordered roughly by mathematical importance and by the likelihood that the existing level-seven structure can be used. Each item is followed, where it applies, by its relation to the items above.

R1. Prove the Lucas-congruence conjecture for A279619. The OEIS entry conjectures Lucas congruences for primes $p \equiv 0, 1, 2, 4 \pmod{7}$. A promising route is to combine a diagonal or constant-term representation with the Cartier operator and Dwork-style unit-root arguments. The exceptional residue classes should be explained geometrically rather than checked case by case. (Same target as Section 13.6; different suggested route.)

- R2. Find a direct binomial or multisum formula for a_n .** The square coefficients t_n have the compact sum (37), while the square-root coefficients are presently most natural through their recurrence or modular expansion. A positive multisum would likely clarify integrality, congruences, and combinatorial meaning simultaneously.
- R3. Determine supercongruences and the underlying motive.** A183204 belongs to the sporadic Apéry-like family [9]. The square-root sequence may encode a lower-weight or half-integral structure. Identify the relevant crystalline Frobenius polynomial and test whether a_{mp^r} satisfies congruences stronger than Lucas congruence.
- R4. Classify all CM evaluations of $A(X_7(\tau))$.** For imaginary quadratic τ , Chowla–Selberg expresses eta values in gamma products. Determine which CM points give algebraic $X_7(\tau)$, then tabulate exact series evaluations $\sum a_n X_7(\tau)^n$. The values at $1/27$ and $1/125$ should be the first entries of a larger level-seven atlas.
- R5. Compute exact connection matrices.** Equation (17) has singular points $0, 1/27, -1$, and ∞ . The present article computes one connection coefficient between 0 and $1/27$. Determine the full 2×2 connection matrices between Frobenius bases at all singular points, preferably in closed gamma form. (Manuscript 58’s companion adds the exact recessive amplitude $D_{\min}$ of (88), a connection datum of the recurrence at the root -1 ; the full matrices remain open.)
- R6. Establish global log-convexity or stronger total positivity.** We proved eventual log-convexity. Exact computation suggests stronger positivity phenomena. Seek a Stieltjes-moment representation, a continued fraction, or a production matrix proving log-convexity for all n and perhaps total positivity of the Hankel matrix (a_{i+j}) . (Overlaps “Positive integral representations” above.)
- R7. Develop rigorous optimal truncation and exponentially small terms.** The algebraic expansion is controlled by the singularity at $1/27$. After optimal truncation, the next singularity at -1 should generate an exponentially small, alternating contribution. Determine the associated transseries and Stokes constants. (Same target as manuscript 58’s late-order and exponentially improved items above. Manuscript 58’s Theorem 2.6 constructs the alternating recessive solution of the recurrence exactly; what remains open is the canonical summation and the Stokes data of a_n itself.)
- R8. Invert the full expansion.** Extend (94) to a systematic inverse transseries for the integer threshold $N(Y)$, including rational corrections and a proof of a uniform rounding rule. (Re-scoped: the systematic inverse with rational corrections is (95)–(97), an instance of `t2:thm:balanced-inverse`, and the rounding brackets are Proposition 11.2. A *uniform* rounding rule is not available: Proposition 11.2 decides $N(Y)$ only when the two ceilings coincide, and manuscript 58 makes no claim that $N(Y) = \lceil x_M(Y) \rceil$ uniformly. What remains open is explicit K_M, Y_M , Section 13.1.)
- R9. Formalize the algebraic core.** A realistic Lean project would formalize the recurrence-to-ODE step, the Frobenius coefficient recurrence, the formal identity $\mathcal{E}_{27}[S] = 0$, and finite-order transfer algebra. The Dedekind eta and Chowla–Selberg inputs would initially be isolated as analytic theorems, making the boundary between formalized algebra and classical complex analysis explicit. (Same programme as Section 13.12.)
- R10. Audit neighboring level-seven OEIS entries.** The same thesis introduced the quartic-recurrence sequence A279613 and related modular functions. Its oscillatory asymptotic is still listed conjecturally. The exact-CM/connection strategy used here suggests a systematic attack on its amplitude and phase constants, although the multiple dominant singularities make the problem substantially harder.

14 Scope, audits, and conclusion

14.1 Manuscript 56’s scope and novelty audit

The mathematical proof of Section 6 uses published structural identities, but the closed form (3) is obtained there by combining them in a way not stated in the inspected sources. The audit performed for manuscript 56 covered:

- the current OEIS pages for A279619 and A183204;
- O’Brien’s 2016 thesis, including its recurrence, modular parametrization, square relation, and numerical asymptotic work;
- Hirschhorn’s proof of the A183204 asymptotic;
- van der Poorten and Williams’ explicit eta-value formula;
- targeted web and repository searches for A279619 together with the gamma product or the displayed decimal constant;
- the existing OEIS-asymptotics reports in the ProveIt repository.

No prior exact identification of the A279619 constant was found in that search: the constant appears not to have been recorded in the OEIS entry, O’Brien’s thesis, Hirschhorn’s paper, or the targeted searches. This is evidence of novelty, not a substitute for a full bibliographic review or peer review, and not a claim that no equivalent formula can exist elsewhere. All theorem statements are supported by the derivations given and by the reproducible checks shipped with the source. Manuscript 58’s audit (Section 1 and 58-gamma-SOURCE_AUDIT.md) reached the same conclusion independently.

14.2 Integration roadmap and conclusion

A natural repository layout separates four components: sequence recurrences and exact certificates; local analytic representation; asymptotic coefficient calculus; and inverse expansions. The first and third components can be tested independently of modular-form machinery. The second supplies the analytic existence and connection constants that formal manipulations alone cannot establish. The fourth should retain an explicit distinction between a continuous inverse scale and an integer threshold function.

No existing Lean declaration is assumed to certify the special-function steps of this report. A useful first integration milestone is a machine-checked discrete Wronskian with rational bracketing of L . The next is a generic formal-series lemma giving (75); both are substantially smaller than a full formalization of the gamma constant. The supplied Python verification programs are a reproducible algebraic audit and a source of test data, not a replacement for that formalization.

The principal outcome is a precise resolution of the asymptotic statement displayed for A279619:

$$a_n \sim \frac{\sqrt{3\pi}}{\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)} \frac{27^n}{n^{3/2}}.$$

The same argument supplies the boundary sum, every algebraic correction with a fixed-order error bound, and a useful convolution calculus. The independent companion construction reveals an alternating recessive solution with an exact gamma-product amplitude. Inversion then yields a Lambert- W scale and rigorous

rounding brackets. The remaining questions concern sharper effective bounds, canonical exponentially improved expansions, arithmetic properties, and broader families, rather than the existence or value of the leading asymptotic constant proved here.

15 Provenance of this report

- **Two manuscripts**, both of batch 73O2 of ProveIt’s incoming-reports intake, both dated October 1, 2026:
 - manuscript 58, *An exact gamma constant for an Apéry-like sequence* (author line “Research report prepared for Vladimir Reshetnikov”; 22-page PDF), archive `A279619_Gamma_Asymptotics.zip`, arrived in commit `aa43cc555`; pin `a5ad4ac2e2d4e642a6c015836a70492d40fcb71` (the repository snapshot it inspected). It is the base text: everything not marked otherwise.
 - manuscript 56, *Exact asymptotics of OEIS A279619: a level-7 CM evaluation, the missing connection constant, and a full coefficient expansion* (author line “A mathematical analysis prepared for Vladimir Reshetnikov”; 15-page PDF), archive `A279619_exact_asymptotics.zip`, arrived in commit `58dbe2cd3`; no pin (it cites no repository revision). It contributes the material listed in Section 1.3.
- **Placement:** commit `6e193dd4f` staged manuscript 58’s article, README and source audit and both manuscripts’ code and data (prefixes `58-gamma-` and `56-exact-`); manuscript 56’s article and README and both PDFs were not staged and survive in their arrival commits.
- **Independence.** Manuscript 58’s pin (12:35 PDT) is earlier than manuscript 56’s arrival (12:56 PDT); manuscript 56 cites no revision. Neither saw the other. Their texts share formulas, the opening terms and the O’Brien bibliography entry, but no prose.
- **Where the merge chose.** (i) Manuscript 58’s analytic justification (the dented disk) is used for both proofs; manuscript 56’s transfer step is kept verbatim in a merge note (Section 7.2). (ii) Results proved in both are printed once in manuscript 58’s wording (the main theorem, $A(\rho)$, the differential equation, the Frobenius recurrence, the formal identity at infinity, the Lambert- W core, the A183204 constant), with manuscript 56’s distinct proofs and forms kept. (iii) Manuscript 56’s notation was rewritten in manuscript 58’s (Section 1.4). (iv) Both abstracts were replaced by one; both title pages by one. (v) Manuscript 56’s reproducibility appendix, which lists its delivery file names, is replaced by the report README. (vi) The two OEIS proposals are both printed (Appendix C); manuscript 56’s gains the credit to O’Brien that manuscript 58’s already had. (vii) The two bibliographies are merged; five entries that manuscript 56 listed but never cited (Cooper, Cox, Apostol, DLMF, Flajolet–Sedgewick) are now cited where they are used.
- [Write note, 2026-10-02, batch 77P5: **A third manuscript**, added in batch 77P5: manuscript 32 of batch 77, *The exact asymptotic constant for A279619 and its fixed weight family* (dated October 2, 2026; empty author line; 15-page PDF), archive `level7-asymptotics-replay.zip`, arrived in commit `096ee7b87` and placed in commit `4f11bc9c0`. It has no pin: it names no repository revision, and it cites neither this report nor manuscripts 56 and 58. It shares 11 of its 445 long source lines with manuscript 58’s source and 19 with manuscript 56’s (formulas and bibliography), and is not a version of either. The placement staged only its replay program (`code/77-32-level7-replay.py`) and that program’s reference output (`data/77-32-level7-replay_results.json`). Its article, PDF, README, build script, manifest (8 of 8 entries verified at placement and retired) and manifest checker were not staged, and its `requirements.txt` is byte-identical to `data/58-gamma-requirements.txt`; all survive in

the arrival commit. It contributes the third route of Section 16 and Remarks 10.2 and 11.3. **Where this addition chose.** (i) Its material is a new last section rather than a section after Section 6, so that no existing number changes; its equations are numbered within that section. (ii) Results it shares with this report (the main theorem, $c_1, \dots, c_6$, the differential equation, the triangular algorithm, the convolution theorem, the A183204 case, the Lambert- W core, the rounding brackets for $m = 1$) are printed once, by reference; its different proofs are kept. (iii) Its notation is rewritten in this report's (Table 7). (iv) Two sentences of its Section 1.1 that present O'Brien's Question 10.2 as answered there and the fixed-weight generators as its contribution are not carried; they are replaced by the scope statement of Section 16. (v) Its abstract and title page are replaced by this report's; its reproducibility section by the report README.]

16 A third derivation of the constant (manuscript 32, batch 77P5)

[Write note, 2026-10-02, batch 77P5: This section is manuscript 32 of batch 77P5 rewritten in this report's notation. It is a third derivation of the constant first proved in Sections 5 and 6 (manuscripts 58 and 56). Manuscript 32 was written without knowledge of either; it presents its proof as answering O'Brien's Question 10.2 [3], which asks for the constant in familiar terms, and its fixed-weight generators as part of its contribution. Relative to this report both are earlier: Theorem 2.1 already answers Question 10.2, and Theorem 10.1 already gives the fixed-weight family to every order. The family theorem is Theorem 10.1; the explicit generator $\mathcal{R}_{m,j}$ of Remark 10.2 is the addition. Equations in this section are numbered within it.]

What is new to the report is the route. (i) The singular coefficient B is computed directly at the Fricke fixed point $\tau_* = i/\sqrt{7}$ from the Fricke involution of the weight-one theta series (Proposition 16.2); the first proof uses a hypergeometric pullback and Ramanujan's series, the second the square sequence A183204. (ii) The value $\eta(i\sqrt{7})^2$ is derived from an unnormalized Epstein zeta function, with an explicit conductor-two factor (Theorem 16.6); the second proof cites the Chowla–Selberg formula. (iii) The Fricke computation is identified as the level-seven case of Guillera and Zudilin's modular limit formula. (iv) In Remarks 10.2 and 11.3: an explicit generator for the convolution coefficients, rounding brackets for every fixed weight, and an implicit-function proof of the inverse expansion. The numerical values in manuscript 32 are checks on the exact expressions, not inputs to their proofs.

16.1 Notation

Manuscript 32 reuses many letters that this report already gives other meanings. Table 7 maps its notation to the one used here; the last column names the tempting false reading. No normalization changes: its $B, C, \lambda, \beta, \ell_j$ and ρ are this report's, and its odd Frobenius series is V with $V(0) = 1$.

Table 7: Notation of manuscript 32 and of this section.

Object	Here	Manuscript 32	Caution
generating-function variable	z	x	—
local variable $1 - 27z$	s	t	—
local coordinate $\sqrt{1 - 27z}$	$\sqrt{s}$	s	32's s is a square root
Fricke fixed point $i/\sqrt{7}$	τ_*	τ_0	here $\tau_0 = (1 + i\sqrt{7})/2$
the point $(1 + i\sqrt{7})/2$	τ_0	α	here α is an exponent
weight-one theta series	$z_7(\tau)$	$\Theta(\tau)$	—
Hauptmodul and its pullback	w_7, X_7	w, X	here w is 58's parameter

Table 7, continued

Object	Here	Manuscript 32	Caution
norm form $j_1^2 + j_1 j_2 + 2j_2^2$	$Q_7(j)$	$Q(m, n)$	here $Q(m) = 1 - m + m^2$
its Gram matrix	$\mathbf{M}$	H	here H is Clausen's ${}_3F_2$
boundary value $A(\rho) = z_7(\tau_*)$	$A_\star$	Θ_0, A_0	—
coefficients of a_n	c_j	p_m	—
formal series at infinity	$S_{27}(u)$	$P(u)$	32 also has $P(x) = (1+x)(1-27x)$
$(1-27z)(1+z)$	$D(z)$	$P(x)$	—
analytic Frobenius series	$\widehat{U} = U/A_\star$	$E(t)$	here E_y is Epstein's series
singular Frobenius series	V	$O(t)$	—
gamma-ratio coefficients	$h_i(a, b)$	$g_m(a, b)$	here g_M is a smooth model
convolution weight	m	k	—
$[z^n]A(z)^m$	$a_n^{(m)}$	$a_{k,n}$	—
its constant $mA_\star^{m-1}C$	$C^{(m)}$	C_k	—
its corrections	$c_i^{(m)}$	$p_{k,m}$	—
$B^2/A_\star^2 = 256\pi^6/(3\mathcal{G}_7^4)$	$\mathbf{q}$	q_* (replay: $\mathbf{q0}$)	here $q_0 = 64/85^3$
generator	$\mathcal{R}_{m,j}$	$R_{k,j}$	here R_n is a scaled term
smooth model of $a_n^{(m)}$	$\mathcal{F}_M^{(m)}$	$F_{k,M}$	here F is a ${}_2F_1$
its truncated series	$S_M^{(m)}$	$P_{k,M}$	—
its large inverse	$\tilde{x}_M^{(m)}(Y)$	$v_{k,M}(Y)$	here $v = N + 1$ (tail)
inverse coefficients	$q_j^{(m)}$	d_j	here d_j are recessive
integer threshold	$N^{(m)}(Y)$	$N_k(Y)$	$N^{(1)} = N$
Epstein argument	σ	z	here z is the GF variable
conductor-two factor	$\varphi_2(\sigma)$	$f(z)$	—
character $(\cdot/7)$ and its L -function	$\chi_{-7}, L(\sigma, \chi_{-7})$	$\chi, L(z, \chi)$	here L alone is the companion limit
$\Gamma(3/7)\Gamma(5/7)\Gamma(6/7)$	$\mathcal{G}_7^-$	$\mathcal{G}_-$	—
Ramanujan's class invariant	$\mathbf{G}_7$	G_7	not the gamma product $\mathcal{G}_7$

16.2 The Fricke fixed point and the local coefficient

Write $Q_7(j) = j_1^2 + j_1 j_2 + 2j_2^2$ for $j = (j_1, j_2) \in \mathbb{Z}^2$, and

$$z_7(\tau) = \sum_{j \in \mathbb{Z}^2} e^{2\pi i \tau Q_7(j)}, \quad \Im \tau > 0. \quad (16.1)$$

It has weight one and the quadratic character of discriminant -7 . With w_7 and X_7 as in (39)–(41), the modular inputs are O'Brien's published identities [3, Theorems 3.7, 3.8, 3.10, 4.6 and 6.1]:

$$A(X_7(\tau)) = z_7(\tau), \quad \frac{dw_7}{d\tau} = 2\pi i w_7(\tau) z_7(\tau)^2, \quad X_7(\tau) = \left(\frac{\eta(\tau)\eta(7\tau)}{z_7(\tau)} \right)^3. \quad (16.2)$$

(The first and third are (41), which identifies z_7 with this theta series.) They identify germs at the cusp, where $X_7 = e^{2\pi i \tau} + O(e^{4\pi i \tau})$; the germs are then continued along an explicit path. A global inverse of X_7 is neither needed nor asserted.

The branch. By (51), $\eta(\tau_*) = 7^{1/4}\eta(i\sqrt{7})$, $w_7(\tau_*) = 1/7$ and $X_7(\tau_*) = \rho$; all eta and theta values below are positive. For $\tau = iy$ with $y \geq 1/\sqrt{7}$, (16.2) gives $\frac{d}{dy}w_7(iy) = -2\pi w_7(iy)z_7(iy)^2 < 0$, so w_7 increases

from 0 to $1/7$ as y decreases from ∞ to $1/\sqrt{7}$; and

$$\frac{dX_7}{dw_7} = \frac{1 - 49w_7^2}{(1 + 13w_7 + 49w_7^2)^2} > 0 \quad (0 < w_7 < 1/7).$$

So X_7 increases from 0 to ρ along the path, its inverse is regular in the interior, and (16.2) identifies the branch reached from the generating function at zero:

$$\lim_{z \uparrow \rho} A(z) = z_7(\tau_*) = A_\star. \quad (16.3)$$

This makes explicit the branch step behind (53).

Proposition 16.1 (Fricke involution). $z_7(-1/(7\tau)) = -i\sqrt{7}\tau z_7(\tau)$.

Proof. The Gram matrix of $2Q_7$ is $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$, with $\det \mathbf{M} = 7$ and $7\mathbf{M}^{-1} = \begin{pmatrix} 4 & -1 \\ -1 & 2 \end{pmatrix}$, which is integrally equivalent to $\mathbf{M}$ (exchange the coordinates and change the sign of one). The two-dimensional Poisson formula for $\sum_{v \in \mathbb{Z}^2} \exp(\pi i \zeta v^\top \mathbf{M} v)$ transforms it, at argument $-1/\zeta$, into the sum with matrix $\mathbf{M}^{-1}$ times the factor $(-i\zeta)/\sqrt{\det \mathbf{M}}$. Taking $\zeta = 7\tau$ and using the integral equivalence gives the identity exactly; no unspecified multiplier remains. $\square$

Proposition 16.2 (The local coefficient at the Fricke point). *Let $\sqrt{s} = \sqrt{1 - 27X_7}$, positive on the real path. Then*

$$\sqrt{s} = \frac{1 - 7w_7}{\sqrt{1 + 13w_7 + 49w_7^2}}, \quad \left. \frac{d\sqrt{s}}{d\tau} \right|_{\tau_*} = -2\pi i A_\star^2 \sqrt{\frac{7}{27}} \neq 0, \quad (16.4)$$

and the singular coefficient of (19) is

$$B = \left. \frac{dz_7}{d\sqrt{s}} \right|_{\sqrt{s}=0} = -\frac{3\sqrt{3}}{4\pi A_\star} < 0. \quad (16.5)$$

Equivalently $CA_\star = 3\sqrt{3}/(8\pi^{3/2})$, which is (32) of the first proof and (66) of the second.

Proof. The algebraic identity for $\sqrt{s}$ follows on squaring, since $1 + 13w + 49w^2 - 27w = (1 - 7w)^2$; both square roots are positive for $0 < z < \rho$. Differentiating Proposition 16.1 at τ_* , where $d(-1/(7\tau))/d\tau = -1$ and $-i\sqrt{7}\tau_* = 1$, gives $-z'_7(\tau_*) = -i\sqrt{7}A_\star + z'_7(\tau_*)$, so $z'_7(\tau_*) = \frac{i\sqrt{7}}{2}A_\star$. By (16.2), $w'_7(\tau_*) = \frac{2\pi i}{7}A_\star^2$; since $1 - 7w_7$ vanishes at τ_* , $d\sqrt{s}/dw_7 = -7\sqrt{7/27}$ there, which gives (16.4). The inverse-function theorem lets z_7 be expanded analytically in $\sqrt{s}$, and $B = z'_7(\tau_*)/(d\sqrt{s}/d\tau)(\tau_*)$ is (16.5). Thus the branch of (16.3) satisfies the convergent local relation $A(z) = A_\star + B(1 - 27z)^{1/2} + O(1 - 27z)$. This fixes the sign and the magnitude of the singular term without an asymptotic formula for A^2 and without a numerical fit. The product form follows from $C = -B/(2\sqrt{\pi})$, (29). $\square$

Remark 16.3 (Guillera–Zudilin). For $Z = A^2$ the local expansion gives

$$\lim_{z \uparrow \rho} \sqrt{D(z)} z Z'(z) = -A_\star B \sqrt{\frac{28}{27}} = \frac{\sqrt{7}}{2\pi}, \quad (16.6)$$

which is exactly the level-seven case of Guillera and Zudilin's equation (27), consistent with their Example 5 and equation (23) [18]. The general connection mechanism and this weight-two limit are prior results; the Fricke computation above is an explicit instance of that mechanism, and the remaining task is the weight-one normalization. In this report's terms the left side is $2A_\star$ times the limiting flux of (20), which Section 3 identifies as $-\frac{1}{2}\sqrt{28/27}B$.

[Write note, 2026-10-02, batch 77P5: Reciprocal note. The report cooper-level-15-moving-zeros in this directory (batch 77, manuscript 33) applies the same Fricke-point connection mechanism (its theorem “A specified real connection”) to Cooper’s sporadic families of levels 11, 14, 15 and 24, proving the ten constants of Cooper’s Table 8, and credits the same Guillera–Zudilin equation (27). The two reports share no theorem, and neither manuscript cites the other; that report already points here.]

16.3 The CM period from an Epstein zeta function

It remains to evaluate $A_\star$. The result is classical in CM theory and is not claimed as new; the purpose is a self-contained derivation in a fixed normalization, because a normalized instead of an unnormalized Epstein zeta function would introduce a spurious power of 7.

Proposition 16.4 (Kronecker limit, derivative form). *For $y > 0$ and $\Re \sigma > 1$ let $E_y(\sigma) = \sum_{(j_1, j_2) \neq (0,0)} (j_1^2 + y^2 j_2^2)^{-\sigma}$. Then E_y continues meromorphically to a neighbourhood of $\sigma = 0$, and*

$$E'_y(0) = -2 \log(2\pi) - 4 \log \eta(iy). \quad (16.7)$$

Proof. Separate $j_2 = 0$, apply the Mellin identity to the remaining terms, and use $\sum_{j \in \mathbb{Z}} e^{-j^2 t} = \sqrt{\pi/t} \sum_{r \in \mathbb{Z}} e^{-\pi^2 r^2/t}$; absolute convergence for $\Re \sigma > 1$ permits the interchanges. The zero frequency gives a gamma integral, and the nonzero frequencies are evaluated by $\int_0^\infty t^{\nu-1} e^{-at-b/t} dt = 2(b/a)^{\nu/2} K_\nu(2\sqrt{ab})$ ($a, b > 0$). The result,

$$\begin{aligned} E_y(\sigma) &= 2\zeta(2\sigma) + \frac{2\sqrt{\pi} \Gamma(\sigma - \frac{1}{2})}{\Gamma(\sigma)} y^{1-2\sigma} \zeta(2\sigma - 1) \\ &\quad + \frac{8\pi^\sigma}{\Gamma(\sigma)} y^{1/2-\sigma} \sum_{n,r \geq 1} n^{1/2-\sigma} r^{\sigma-1/2} K_{\sigma-1/2}(2\pi ynr), \end{aligned} \quad (16.8)$$

gives the continuation; the double Bessel series and its derivatives in the order converge locally uniformly by exponential decay, so termwise differentiation is legitimate. At $\sigma = 0$ the first term contributes $4\zeta'(0) = -2 \log(2\pi)$; since $1/\Gamma(\sigma) = \sigma + O(\sigma^2)$, $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$ and $\zeta(-1) = -\frac{1}{12}$, the second contributes $\pi y/3$; and $K_{-1/2}(u) = \sqrt{\pi/(2u)} e^{-u}$ makes the third contribute $4 \sum_{n,r \geq 1} e^{-2\pi ynr}/r = -4 \sum_{n \geq 1} \log(1 - e^{-2\pi yn})$. With the product for η the three combine to (16.7). $\square$

In the determinant-one normalization $\tilde{E}_y(\sigma) = y^\sigma E_y(\sigma)$ one has instead $\tilde{E}'_y(0) = -2 \log(2\pi) - \log y - 4 \log \eta(iy)$, because $E_y(0) = -1$; the extra $-\log y$ disappears when the normalization is undone. This agrees with the classical limit-formula literature [21]; the derivation makes the convention independent of a citation.

Proposition 16.5 (The conductor-two factor). *Let $\chi_{-7}(a) = (a/7)$ and $\mathcal{O} = \mathbb{Z}[\tau_0]$, $\tau_0 = (1 + i\sqrt{7})/2$. For $\Re \sigma > 1$,*

$$\sum_{j \neq 0} Q_7(j)^{-\sigma} = 2\zeta(\sigma) L(\sigma, \chi_{-7}), \quad E_{\sqrt{7}}(\sigma) = 2\varphi_2(\sigma) \zeta(\sigma) L(\sigma, \chi_{-7}), \quad (16.9)$$

with $\varphi_2(\sigma) = 1 - 2^{1-\sigma} + 2^{1-2\sigma}$, and both identities continue meromorphically.

Proof. $\mathcal{O}$ is the maximal order of discriminant -7 . Minkowski’s bound $2\sqrt{7}/\pi < 2$ puts an integral ideal of norm one in every class, so the class number is one, and the only units are ± 1 . Counting a generator and its negative for each nonzero principal ideal gives the first identity (equivalently, compare Euler factors at split, inert and ramified primes). The form $j_1^2 + 7j_2^2$ belongs to the suborder $\mathbb{Z}[i\sqrt{7}] = \mathbb{Z} + 2\mathbb{Z}\tau_0$. Modulo 2 the minimal polynomial $T^2 - T + 2$ becomes $T(T - 1)$, so $\mathcal{O}/2\mathcal{O} \simeq \mathbb{F}_2 \times \mathbb{F}_2$, and the suborder consists of

the elements with equal residues in the two factors. A nonzero such element has residues $(1, 1)$, hence odd norm, or $(0, 0)$, hence lies in $2\mathcal{O}$. The prime 2 splits into two primes of norm 2; excluding both multiplies the ideal sum by $(1 - 2^{-\sigma})^2$ and counts exactly the odd-norm elements, while multiplication by 2 multiplies norms by 4 and counts the other class. The two sets are disjoint, which gives $\varphi_2 = (1 - 2^{-\sigma})^2 + 4^{-\sigma}$. The identity holds first for $\Re \sigma > 1$ and then by continuation. No CM formula for a nonfundamental discriminant is assumed. $\square$

Theorem 16.6 (The eta value; classical).

$$\eta(i\sqrt{7})^2 = \frac{\mathcal{G}_7}{8 \cdot 7^{1/4} \pi^2}, \quad A_\star = 3\eta(\tau_*)\eta(i\sqrt{7}) = 3 \cdot 7^{1/4} \eta(i\sqrt{7})^2 = \frac{3\mathcal{G}_7}{8\pi^2}, \quad B = -\frac{2\pi\sqrt{3}}{\mathcal{G}_7}. \quad (16.10)$$

Proof. From $L(\sigma, \chi_{-7}) = 7^{-\sigma} \sum_{a=1}^6 \chi_{-7}(a) \zeta(\sigma, a/7)$, $\zeta(0, u) = \frac{1}{2} - u$ and $\zeta'(0, u) = \log \Gamma(u) - \frac{1}{2} \log(2\pi)$ (the character is +1 at 1, 2, 4 and -1 at 3, 5, 6),

$$L(0, \chi_{-7}) = 1, \quad L'(0, \chi_{-7}) = -\log 7 + \log(\mathcal{G}_7/\mathcal{G}_7^-).$$

With $\varphi_2(0) = 1$, $\varphi_2'(0) = -2 \log 2$, $\zeta(0) = -\frac{1}{2}$ and $\zeta'(0) = -\frac{1}{2} \log(2\pi)$, differentiating (16.9) gives $E'_{\sqrt{7}}(0) = \log(14/\pi) - \log(\mathcal{G}_7/\mathcal{G}_7^-)$. Comparing with (16.7) at $y = \sqrt{7}$ and using Gauss's multiplication formula $\mathcal{G}_7 \mathcal{G}_7^- = \prod_{a=1}^6 \Gamma(a/7) = (2\pi)^3 / \sqrt{7}$ gives $\eta(i\sqrt{7})^4 = \mathcal{G}_7^2 / (64\sqrt{7} \pi^4)$, and positivity fixes the square root. At τ_* the third identity of (16.2) and positivity give $A_\star = 3\eta(\tau_*)\eta(i\sqrt{7})$, and $\eta(\tau_*) = 7^{1/4} \eta(i\sqrt{7})$. Finally Proposition 16.2 gives B . $\square$

The first value is (60) squared, the second is (62), and the third is (67).

Remark 16.7 (A classical cross-check, and its relation to the second proof). Manuscript 32 compares the value with the fundamental-discriminant formula. Zagier gives [20, p. 95] $7^{1/4} |\eta(\tau_0)|^2 = \mathcal{G}_7 / (4\pi^2)$. In Rebák's notation the class invariant is $\mathbf{G}_n = 2^{-1/4} r^{-1/24} \prod_{j \geq 0} (1 + r^{2j+1})$ with $r = e^{-\pi\sqrt{n}}$, and the classical evaluation $\mathbf{G}_7 = 2^{1/4}$ is recalled in the proof of his Theorem 4.1 [22]. Dividing the eta products with $r = e^{-\pi\sqrt{7}}$ gives $|\eta(\tau_0)| / \eta(i\sqrt{7}) = r^{-1/24} \prod_{j \geq 0} (1 + r^{2j+1}) = \sqrt{2}$, which again gives (16.10). In substance this is the route of Section 6: Zagier's value is (57) (there from van der Poorten and Williams's form of Chowla–Selberg), and the ratio $\sqrt{2}$ is (60) (there from Weber's function with $u = -1$). Neither classical evaluation is claimed as new. The cross-check separates the discriminant -7 period at τ_0 from the conductor-two point $i\sqrt{7}$.

16.4 Continuation, transfer and the third proof

Manuscript 32 justifies the continuation without the dented disk of Section 3. The equation (17) has finite singular points 0, ρ and -1 only. The germ at zero is analytic, and every other point of $|z| < \rho$ is ordinary, so A is single-valued and analytic there; this also justifies continuing (16.2) along the path above. For $\rho < R < 1$ the slit disk $\{|z| < R\} \setminus [\rho, R)$ contains no singular point other than 0; loops in it minus $\{0\}$ have monodromy generated by a loop about zero, which fixes the analytic germ. So the continued solution is single-valued, its singularity at zero is removable, and it supplies the complex Δ -domain at ρ that singularity analysis needs, not merely a real-axis limit. In $s = 1 - 27z$ the equation is (18), with exponents 0 and $\frac{1}{2}$, so $A = A_\star \widehat{U}(s) + B\sqrt{s} V(s)$ with no logarithmic term; the constants are fixed by (16.3) and Theorem 16.6, and $B \neq 0$ makes ρ a genuine singularity, the only one on $|z| = \rho$.

For the transfer, expand $\widehat{U}$ through degree $M + 1$ and V through degree M . The subtracted integral-power part is a polynomial in z ; it is not necessary to assert that the full analytic branch $U(1 - 27z)$ extends to $z = 0$. The remainder is analytic in the slit domain and of local size $O(s^{M+3/2})$, so its coefficients are

$O_M(27^n n^{-M-5/2})$ by a Hankel-type contour near ρ and an outer contour beyond ρ [12, Chapter VI]. With (73) this gives

$$a_n = 27^n B \sum_{j=0}^M v_j \frac{\Gamma(n-j-\frac{1}{2})}{\Gamma(-j-\frac{1}{2})\Gamma(n+1)} + O_M(27^n n^{-M-5/2}), \quad (16.11)$$

whose leading term is $-B/(2\sqrt{\pi}) = \sqrt{3}\pi/\mathcal{G}_7 = C$. The leading bound makes $\sum a_n \rho^n$ convergent, and Abel's theorem with (16.3) identifies its sum as $A_\star$. This completes the third proof of Theorem 2.1.

16.5 Generators for every fixed order

Both Frobenius series come from one recurrence. For $r \in \{0, \frac{1}{2}\}$ put $f_{r,-1} = 0$, $f_{r,0} = 1$; then for $j \geq 1$ and $v = r + j$,

$$f_{r,j} = \frac{(29(v-1)^2 + 14(v-1) + \frac{20}{9})f_{r,j-1} - (v-\frac{4}{3})(v-\frac{5}{3})f_{r,j-2}}{14v(2v-1)}, \quad (16.12)$$

with no vanishing denominator. For $r = \frac{1}{2}$ this is (69) ($f_{1/2,j} = v_j$). For $r = 0$ it gives the normalized analytic part, which the proof of Theorem 10.1 uses only through its first coefficient:

$$\widehat{U}(s) = \frac{U(s)}{A_\star} = 1 + \frac{10}{63}s + \frac{986}{11907}s^2 + \frac{89984}{1607445}s^3 + O(s^4). \quad (16.13)$$

Both converge near zero by Frobenius theory, not merely as formal series.

Let B_k be the Bernoulli polynomials and define rational polynomials $h_i(a, b)$ by

$$\sum_{i \geq 0} h_i(a, b) u^i = \exp\left(\sum_{r \geq 1} \frac{(-1)^{r+1}(B_{r+1}(a) - B_{r+1}(b))}{r(r+1)} u^r\right), \quad (16.14)$$

which is the fixed-order expansion $\Gamma(n+a)/\Gamma(n+b) \sim n^{a-b} \sum_i h_i(a, b) n^{-i}$ obtained by subtracting two Stirling series for $\log \Gamma$ [13, §5.11]. Since $\Gamma(-\frac{1}{2})/\Gamma(-j-\frac{1}{2}) = (-1)^j (\frac{3}{2})_j$, (16.11) gives

$$c_i = \sum_{j=0}^i v_j (-1)^j (\frac{3}{2})_j h_{i-j}(-j-\frac{1}{2}, 1), \quad (16.15)$$

each coefficient by finitely many rational operations. This is the ‘‘Bernoulli-polynomial gamma-ratio’’ route by which the batch-73O2 intake checked $c_0, \dots, c_8$. Manuscript 32's second, independent generator is the formal equation $\mathcal{E}_{27}[S_{27}] = 0$ of (74) with the triangular step (75) (denominator $28m$), proved in both earlier manuscripts and printed once there; it gives uniqueness and rationality and checks (16.15). The formal equation identifies the coefficients of an actual expansion only because the expansion has already been established; a formal ansatz alone determines neither the connection constant nor the remainder. Manuscript 32 prints $c_1, \dots, c_6$, equal to (5) and (76).

16.6 Scope, prior work and the replay

Manuscript 32's statement of prior work and scope, kept here:

- O'Brien's thesis establishes the recurrence and the modular parametrization; its Conjecture 8.1 gives the corrections c_1, c_2, c_3 , its Section 8.4.3 estimates the leading constant numerically, and its Question 10.2 asks for an exact expression [3]. The OEIS entry inspected on October 2, 2026 still labelled the leading equivalent conjectural [1].

- The weight-two sequence A183204, its constant and its corrections through the sixth order are Hirschhorn's [4]. Cooper's 2025 article [19], inspected in its author preprint of May 8, 2024, recovers this known weight-two asymptotic in its Section 10; its Section 11 discusses the weight-one sequence in connection with congruences and cites O'Brien, but does not supply the weight-one connection constant.
- The general modular radial-connection mechanism predates the manuscript (Remark 16.3); its role here is to connect the weight-one amplitude to the CM value $A_\star$.
- The source search behind these comparisons is bounded and does not establish exhaustive priority. The modular identities, the classical CM values, Poisson summation, Frobenius theory, singularity transfer and implicit inversion are established tools.
- All asymptotic statements have a fixed weight m and a fixed truncation order, and the implied constants may depend on both. No claim is made about a growing weight, a growing truncation order, uniformity in either, effective inverse constants or cutoffs, exponentially improved expansions, optimal truncation, or the oscillatory companion A279613.

The replay program `code/77-32-level7-replay.py` uses exact rational arithmetic for the Frobenius, recurrence, family and inverse generators and high-precision floating point for the numerical checks; its reference output is `data/77-32-level7-replay_results.json` (order 6, $n \leq 1600$, 100 digits). It compares the two rational generators, checks the displayed coefficients, generates exact a_n from (1) with integrality and monotonicity checked through $n = 1600$, checks the weight-two binomial sum for $n \leq 40$, evaluates the eta, theta, Fricke and Epstein identities to relative error below 10^{-80} at 100 digits, and compares forward and inverse residuals for $m = 1, 2, 3$; for instance $n(a_n/(C27^n n^{-3/2}) - 1) \rightarrow -215/1008$, and multiplying by n^{M+1} after subtracting M corrections exposes the next coefficient. These are finite symbolic identities and numerical consistency checks. They do not replace the analytic proof, establish integrality for all indices, prove interval bounds, or turn the existential inverse brackets into a certified rounding algorithm. The intake reran it on a copy; the output agreed with the reference in every entry except the elapsed time and the Python version (README).

A Checking the classical period normalization

The table in Zucker's singular-value notes [8, Table 1] uses $K[N] = K(k_N)$ with $K(k'_N)/K(k_N) = \sqrt{N}$. At $N = 7$ it gives

$$K[7] = \frac{2^{5/7} 7^{3/4}}{14} \frac{\beta(1/7)\beta(2/7)}{\beta(1/14)}, \quad k_7^2 = \frac{8 - 3\sqrt{7}}{16}, \quad (103)$$

where $\beta(t) = \Gamma(t)^2/\Gamma(2t)$. We spell out the simplification because confusing the elliptic modulus with its square would give an incorrect constant.

Set $g_1 = \Gamma(1/7)$, $g_2 = \Gamma(2/7)$, $g_4 = \Gamma(4/7)$, and $g_{1/2} = \Gamma(1/14)$. Then

$$\frac{\beta(1/7)\beta(2/7)}{\beta(1/14)} = \frac{g_1^3 g_2}{84 g_{1/2}^2}.$$

The gamma duplication formula at $1/14$ is

$$g_{1/2} g_4 = 2^{6/7} \sqrt{\pi} g_1.$$

It follows that the beta quotient is $g_1 g_2 g_4 / (2^{12/7} \pi) = \mathcal{G}_7 / (2^{12/7} \pi)$. Substitution into (103) gives

$$K[7] = \frac{7^{3/4} \mathcal{G}_7}{28\pi} = \frac{\mathcal{G}_7}{4\pi 7^{1/4}},$$

which is exactly (34). The classical table evaluation is an external input; the algebraic conversion here checks that its normalization matches the one used in the proof.

B Two derivations from manuscript 56

B.1 The recurrence-to-ODE identity

For completeness, write $\vartheta = z d/dz$. Then

$$\sum_{n \geq 0} n a_n z^n = \vartheta A, \quad \sum_{n \geq 0} n^2 a_n z^n = \vartheta^2 A.$$

The left side of (1), summed over $n \geq 0$, is $A' + zA''$. The first term on the right gives $26z^2 A'' + 39zA' + 2A$, and the shifted term gives $27z^3 A'' + 54z^2 A' + 6zA$. Their difference is exactly (17).

B.2 A compact derivation of the coefficient algorithm

Assume

$$a_n = C 27^n n^{-3/2} \mathcal{A}_n, \quad \mathcal{A}_n \sim S_{27}(1/n).$$

Divide (1) by $C 27^n n^{1/2}$ and put $t = 1/n$. The three terms become

$$27(1+t)^{1/2} S_{27}\left(\frac{t}{1+t}\right), \quad (26 + 13t + 2t^2) S_{27}(t), \\ \left(1 - t + \frac{2}{9}t^2\right) (1-t)^{-3/2} S_{27}\left(\frac{t}{1-t}\right),$$

which is $\mathcal{E}_{27}[S_{27}] = 0$ in (74). The coefficient of t^2 is $-215/36 - 28c_1$, giving $c_1 = -215/1008$; successive coefficients proceed triangularly: after the automatic cancellations in degrees 0 and 1, each next coefficient determines one new c_j linearly. This proves that all c_j are rational and supplies a fast exact algorithm.

C Possible OEIS formula updates

Both manuscripts propose an update; neither was submitted. The following is suggested mathematical content for an update after independent review. No edit to OEIS has been made as part of this work.

C.1 Manuscript 58's proposal

Let $G = \Gamma(1/7)\Gamma(2/7)\Gamma(4/7)$. Then

$$a(n) \sim \frac{\sqrt{3\pi}}{G} \frac{27^n}{n^{3/2}}, \quad \sum_{n \geq 0} \frac{a(n)}{27^n} = \frac{3G}{8\pi^2}.$$

More precisely, for every fixed truncation order there is an asymptotic expansion with a remainder of the next order. Its first corrections are

$$1 - \frac{215}{1008n} - \frac{1265}{290304n^2} + \frac{4683055}{877879296n^3} + O(n^{-4}).$$

The first three corrections were already calculated formally by O'Brien [3].

The separate Lucas-congruence statement is not affected by this report.

C.2 Manuscript 56's proposal

A compact statement suitable for adaptation to the OEIS entry after independent review (its plain-text form is `data/56-exact-OEIS_update.txt`):

Let $\mathcal{G}_7 = \Gamma(1/7)\Gamma(2/7)\Gamma(4/7)$. Then

$$a(n) \sim \frac{\sqrt{3\pi}}{\mathcal{G}_7} 27^n n^{-3/2}.$$

More precisely,

$$a(n) = \frac{\sqrt{3\pi}}{\mathcal{G}_7} 27^n n^{-3/2} \left(1 - \frac{215}{1008n} - \frac{1265}{290304n^2} + \frac{4683055}{877879296n^3} + O(n^{-4}) \right).$$

For the generating function $A(x) = \sum_{n \geq 0} a(n)x^n$,

$$A(1/27) = \frac{3\mathcal{G}_7}{8\pi^2}, \quad A(1/125) = \frac{5\mathcal{G}_7}{16\pi^2}.$$

At the dominant singularity,

$$A(x) = \frac{3\mathcal{G}_7}{8\pi^2} - \frac{2\pi\sqrt{3}}{\mathcal{G}_7} \sqrt{1-27x} + O(1-27x).$$

[Merge note, batch 7302: The first three corrections were calculated formally by O'Brien (2016 thesis, §8.4) [3]. Manuscript 56's proposal and its plain-text file do not say so; any submission must add this credit.]

References

- [1] The OEIS Foundation Inc., *A279619: Expansion of the generating function of A002652 in powers of the generating function of A279618*. Entry by Lynette O’Brien, with later contributions; consulted October 1, 2026. The displayed entry revision was August 22, 2026. <https://oeis.org/A279619/internal>.
- [2] The OEIS Foundation Inc., *A183204*, recurrence, differential equation, generating-function relations, and asymptotic formula. Consulted October 1, 2026. <https://oeis.org/A183204/internal>.
- [3] L. A. O’Brien, *Modular forms and two new integer sequences at level 7*, Master of Science thesis, Massey University, 2016. Especially Theorem 6.1 and Section 8.4, pp. 63–67.
<https://oeis.org/A279619/a279619.pdf>;
<https://mro.massey.ac.nz/items/ad1bd663-2b56-4eed-ad1a-f130bb7fe757>.
- [4] M. D. Hirschhorn, *A sequence considered by Shaun Cooper*, New Zealand Journal of Mathematics **43** (2013). <https://web.maths.unsw.edu.au/~mikeh/webpapers/paper182.pdf>.
- [5] A. J. van der Poorten and K. S. Williams, *Values of the Dedekind eta function at quadratic irrationalities*, Canadian Journal of Mathematics **51** (1999), no. 1, 176–224.
<https://doi.org/10.4153/CJM-1999-011-1>.
- [6] R. Hemmecke, P. Paule, and C.-S. Radu, *Computer-assisted construction of Ramanujan–Sato series for 1 over π* , manuscript dated January 31, 2025. Equation (98), p. 32, is the level-seven identity used here.
https://www3.risc.jku.at/publications/download/risc_7134/.RamanujanSatoOneOverPi.pdf.
- [7] J. M. Borwein and I. J. Zucker, “Fast evaluation of the gamma function for small rational fractions using complete elliptic integrals of the first kind,” *IMA Journal of Numerical Analysis* **12** (1992), 519–526.
<https://academic.oup.com/imajna/article-abstract/12/4/519/690281>.
- [8] I. J. Zucker, *Singular value theory*, exercise notes in Jonathan Borwein’s archived materials, Table 1, entry $K[7]$. The notes refer to Borwein and Zucker’s elliptic-integral evaluations; consulted October 1, 2026. <https://carmamaths.org/jon/Preprints/Books/EMA/Exercises/For%20others/K-beta.pdf>.
- [9] S. Cooper, *Sporadic sequences, modular forms and new series for $1/\pi$* , Ramanujan Journal **29** (2012), 163–183.
- [10] D. A. Cox, *Primes of the Form $x^2 + ny^2$* , 2nd ed., Wiley, 2013.
- [11] T. M. Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Springer, 1990.
- [12] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. Chapter VI: singularity analysis and transfer of local expansions.
- [13] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*: Sections 4.13, 5.11, and 15.10 (Lambert’s W function, gamma asymptotics, the hypergeometric differential equation), cited by manuscript 58; Chapters 5, 23 and 27 (gamma function, modular functions, functions of number theory), listed by manuscript 56. Consulted October 1, 2026. <https://dlmf.nist.gov/>.
- [14] H. Cohen and W. Zudilin, *Continued fractions and irrationality measures for Chowla–Selberg gamma quotients*, arXiv:2510.00215, 2025, revised versions through 2026. Related work, not a theorem dependency here. <https://arxiv.org/abs/2510.00215>.

- [15] V. Reshetnikov, *ProveIt*, Analysis/Transseries/README.md, repository snapshot a5ad4ac2e2d4e642a6c015836a70492d40fcbc71.
<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/Transseries>.
- [16] ProveIt, *Combinatorial Transseries and Their Inverses*, Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/, Theorem t2:thm:balanced-inverse (explicit centered inverse coefficients). Cited in the merge; neither manuscript cites it.
- [17] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*, Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/, Theorem p0:thm:staircase (staircase recovery and the separation condition). Cited in the merge.
- [18] J. Guillera and W. Zudilin, *Ramanujan-type formulae for $1/\pi$: the art of translation*, in *The Legacy of Srinivasa Ramanujan*, Ramanujan Mathematical Society Lecture Notes Series **20** (2013), 181–195. Example 5, equation (23) and the general modular limit (27). Cited by manuscript 32.
<https://arxiv.org/abs/1302.0548v2>.
- [19] S. Cooper, *Apéry-like sequences defined by four-term recurrence relations*, Contemporary Mathematics **818** (2025), 137–179. <https://doi.org/10.1090/conm/818/16373>. Manuscript 32 inspected the author preprint arXiv:2302.00757v2 (May 8, 2024): Section 10 treats the known weight-two asymptotic, Section 11 weight-one congruences.
- [20] D. Zagier, *Elliptic modular forms and their applications*, in *The 1-2-3 of Modular Forms*, Universitext, Springer, 2008, 1–103. The Chowla–Selberg discussion and the discriminant -7 period on p. 95. Cited by manuscript 32. https://doi.org/10.1007/978-3-540-74119-0_1.
- [21] F. Chamizo, *A simple evaluation of a theta value, the Kronecker limit formula and a formula of Ramanujan*, The Ramanujan Journal **59** (2022), 947–954. Cited by manuscript 32.
<https://doi.org/10.1007/s11139-022-00586-0>.
- [22] Ö. Rebák, *The three missing terms in Ramanujan’s septic theta function identity*, The Ramanujan Journal **60** (2023), 885–911. Equation (5) and the proof of Theorem 4.1. Cited by manuscript 32.
<https://doi.org/10.1007/s11139-022-00584-2>.